TEMPLATE

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (PDD)

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VERSION **v. 1.1**

RELATED SUPPORT

– **TEMPLATE GUIDE Key Project Information & VPA Design Document v.1.1**

This document contains the following Sections

Key Project Information

KEY PROJECT INFORMATION

GS ID of Project	GS11070
Title of Project	GS1340 Efficient cookstoves in Burkina Faso – VPA-25 – Solidagro F3PA cookstoves in Passoré
Time of First Submission Date	23/03/2021
Date of Design Certification	N/A
Version number of the VPA-DD	5.0
Completion date of version	28/06/2022
Coordinating/managing entity	Association tiipaalga
VPA Implementer (s)	CO2logic Solidagro
Project Participants and any communities involved	Municipalities of Arbole, Bagare, Bokin, Samba, Pilimpikou, Kirsi and Yako in the province of Passoré in the North Region
Host Country (ies)	Burkina Faso
GS ID and Title of applicable Design Certified VPA	N/A
GS ID and Title of applicable Performance Certified VPA	N/A
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☒ Micro scale ☐ Small Scale ☐ Large Scale
Other Requirements applied	PoA microscale
Methodology (ies) applied and version number	Gold Standard Simplified Methodology for Efficient cookstoves, February 2013 v.1.0
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☒ Regular

	☐ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact	Estimated Annual Average	Units or Products
1. No poverty	Average household annual savings	7 €	Amount
	Total estimated amount saved by stove users on wood fuel purchase for the group of VPA's	13,558 €	
3. Good health and well being	Smoke level reduction	>90%	Percentage
	Incidence of coughing reduction	>90%	
	Incidence of respiratory illness reduction	>90%	
	Incidence of itchy eyes reduction	>90%	
4. Quality Education	Number of workshops for women for the group of VPA's	72	Number
	Number of employees provided skill development training for the group of VPA's	5.3	Number
5. Gender Equality	Average time saving associated with cooking time and fuel collection per woman per week	2.2 hours	Hours

	Number of women serving in managerial/ leadership /ownership role for the group of VPA's	1,080	Number
7. Affordable clean energy	Number of households equipped with F3PA cookstoves for the group of VPA's	9,632	Households
13. Climate Action	Emissions reductions	9,427	VERs
15. Life on land	Reduced deforestation attributed to wood fuel savings for the group of VPA's	14,849	Tonnes of non-renewable wood

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

Burkina Faso heavily depends on woodfuels as the main energy source. Having one of the highest annual population increments and lowest indexes of human development anywhere in the world, Burkina's dependency on firewood and charcoal is forecasted to even increase in the coming decades. This increasing woodfuel consumption is believed to be the main driver of deforestation and forest degradation[1]. According to the FAO, during the period between 1990 and 2010, forest cover has declined at an average of 1% per year[2]. About 14% of Burkina Faso's territory belong to protected areas[3] but a rapid intensification of land-use is predicted for the future due to high annual population growth (2,93%[4]) and the developing economy.

The main objective of the VPA is to contribute to mitigating the effects of climate change and reduce poverty among rural households by promoting the use of improved three-stone cookstoves that are more economical in woodfuel in 7 rural municipalities (Arbole, Bagare, Bokin, Samba, Pilimpikou, Kirsi and Yako) in Passoré province in the North region of Burkina Faso.

The vegetation cover of the North region is characterized by steppe, tiger bush, shrubby savannah and wooded savannah[5]. The plant cover of this region is seriously threated as there is a continuous reduction or even disappearance of woody and herbaceous species. Consequently, the desertification of the area is enhanced. This degradation is influenced by different anthropic factors a.o. demographic pressure, bush fires, cultivation and breeding methods.

In the North Region, 93.6%[6] of households use firewood as their main energy source for cooking. In rural Burkina Faso as a whole, this percentage reaches 94.9%[7], which constitutes a strong disparity compared to the urban environment (59%).

The pressure on wood resources for household cooking undeniably contributes to the desertification of the rural landscape in the northern region and accentuates household poverty and food insecurity. Those negative externalities are enhanced by the growing

[1] Arevalo Javier, 2016, Improving woodfuel governance in Burkina Faso: The experts'assessment, Renewable and Sustainable Energy Reviews, page 2 (see document 'arevalo.pdf').
[2] Readiness Preperation Plan for REDD Burkina Faso, Ministry of the Environment and Sustainable Development Page 54 (see document '13-11-08 R-PP Burkina EN VFin20h.pdf')
[3] Phytotaxa, 2017, Diversity, distribution and preliminary conservation status of the flora of Burkina Faso page 3 (see document 'Phytotaxa 2017.pdf')
[4] RGPH-5 Résultats préliminaires (see document : 'Montage Rapport RGPH5 def finaliser_04122020.pdf' pg 17)
[5] Ministry of Mines and Energy, 2013, Support project for the Electricity Sector (PASEL), pages 21-22 (see document 'Pasel_2013').
[6] INSD 2019, Table 2.2.36 : 'Répartition des ménages par région selon le mode d'énergie pour la cuisine en 2014 (en %)', pg 55 (see document 'Annuaire_Statistique_National_2019').
[7] National Institute for Statistics and Demography (INSD), 2007, Analysis of the results of the survey for living conditions of the households, pg 66. (see document 'INSD_2007').

population of the country[8], and more specifically in the North Region. Between 2006[9] and 2019[10], the population in the project area (municipalities of Arbole, Bagare, Bokin, Samba, Pilimpikou, Kirsi and Yako) has increased by 30% and reaches now a total of 361,608 habitants. National statistics show that 25.2%[11] of the inhabitants of the North region can be considered as among the poorest (indicator of non-monetary standard of living). This percentage is the highest at country level.

For customary reasons, most of the households of the North region (94.4%[12]) use traditional three-stones stoves for domestic cooking. This simple but inefficient technology enhances the pressure on wood resources and contributes to the desertification of the landscape. The present project promotes the distribution and utilization of the mud made 3 stones efficient woodstove "F3PA" in the project intervention area. The efficient F3PA cookstove will replace the traditional open air three stone cooking method whilst respecting the local three stone cooking culture. This is possible as the efficient F3PA cookstove, seen on Figure 1, will integrate the three stones from each household inside its design. These three stones represent the pillar of the household's marital union. The F3PA is significantly more efficient than the traditional open fire three stone cooking method[13]. The project will thus help reduce wood consumption by more than half in each household and therefore preserve the local forests and their biodiversity. The F3PA has further benefits such as avoiding hazardous open flame systems and reducing the quantity of harmful smoke in the local rural village households. Local families and women also benefit significantly through a reduction in time spent and distance walked in collecting wood. The project does not consist in a fuel switch as locally available wood is still being used.

The following energy efficiency microscale VPA is part of a group of 4 VPA's. It is proposed by Solidagro, an agro-ecology and climate NGO, the Association for the Protection of the Environment and Rural Development (APEDR) and CO2logic, a Belgian consultancy company for the technical assistance with the carbon market aspects. An external partner of the project is the local Association tiipaalga, which has an extensive experience in the domain of improved cookstove projects in the Burkina Faso more specifically in the provinces of Bam, Loroum, Kourwéogo and Nahouri. Because the association initially developed the F3PA technology, tiipaalga will bring its technical

[8] Fifth General Population and Housing Census of Burkina Faso (RGPH-5), 2019, Résultats préliminaires, pg 17 (see document: 'Montage Rapport RGPH5 def finaliser_04122020.pdf)
[9] Fourth General Population and Housing Census of Burkina Faso (RGPH-4), 2007, pg 48 (see document: 'RGPH_2006')
[10] Fifth General Population and Housing Census of Burkina Faso (RGPH-5), 2019, Résultats préliminaires, pg 50 (see document: 'Montage Rapport RGPH5 def finaliser_04122020.pdf')
[11] National Institute for Statistics and Demography (INSD), 2018, Integrated Regional Survey on Employment and the Informal Sector (ERI-ESI), pg 139 (see document 'rapport_final_burkina_faso').
[12] Monographie de la Région du Nord, Ministère de l'Economie et des Finances, Burkina Faso, Table 12.6, page 131 (see document 'Monographie-Nord RGPH 2006.pdf')
[13] Rapport sur les tests de performances énegétiques des Foyers trois pierres améliorés (F3PA) de l'association Tiipaalga, Laboratoire Biomasse Energie et Biocarburnat de 2IE, Ouagadougou, July 2015 (see document ' tiipaalga_Rapport de tests de performance énergétiques_F3PA_24_07_2015_VF.pdf')

expertise and experience to ensure an optimal implementation and follow-up of the activities.

Figure 1 Locally produced efficient F3PA cookstove

The following energy efficiency microscale VPA is part of a group of 4 VPA's. It is proposed by Solidagro, an agro-ecology and climate NGO, the Association for the Protection of the Environment and Rural Development (APEDR) and CO2logic, a Belgian consultancy company for the technical assistance with the carbon market aspects. An external partner of the project is the local Association tiipaalga, which has an extensive experience in the domain of improved cookstove projects in the Burkina Faso more specifically in the provinces of Bam, Loroum, Kourwéogo and Nahouri. Because the association initially developed the F3PA technology, tiipaalga will bring its technical expertise and experience to ensure an optimal implementation and follow-up of the activities.

The project's approach involves training of leader women (or 'Monitrice Endogène') in the rural zones to build, use and maintain these efficient cookstoves themselves using local material. The training sessions are done by instructors of APEDR trained by Association tiipaalga. They include sensitization on hygienic usage and on the threat of climate change and health hazards related to the old cooking system and inform on the health and environmental benefits of using such efficient cookstoves. These leader women conduct the same training sessions with the women in their villages and help them to build and use in proper manner the efficient F3PA cookstoves on a voluntary basis.

Each F3PA efficient cookstove will be built according to a strict construction protocol established by Association tiipaalga[14]. The efficient cookstove construction instructions are published during the training and all levels of the process will be trained to have full knowledge of this construction criterion. The skills they require to build their own personally fabricated standardized efficient F3PA cookstove and how to use and maintain it. The training sessions also involve a supervisory role to ensure the efficient F3PA cookstoves meet the protocol fabrication standard prior to be given an individual project registry code.

The participating women will sign an agreement in exchange for this training whereby they consent the carbon offsets related to this project to PD who will use these future revenues to cover the costs of the project extension as well as costs related to the staff and training. Each efficient cookstove or an equivalent in-service device will be operational over the full lifespan of the micro-scale VPA project.

The start of the project is considered as the day of installation of the first F3PA in the VPA. The start date of the VPA 25 project is 10/03/2021.

A.1.1. Eligibility of the project under approved PoA

The project will be included in the approved PoA GS1340 Efficient cookstoves in Burkina Faso. The following table resumes the eligibility criteria met by the VPA.

No.	**Eligibility Criterion**	**Description/ Required condition**	**Means of Verification/Supporting evidence for inclusion**
1	Technological requirements	The VPA consists of the implementation or distribution of single pot or multi pot portable or an in-situ wood burning cookstoves with a specified efficiency of at least 20% to meet thermal energy requirements for household cooking as per Gold Standard Simplified Methodology for Efficient Cookstoves.	The project aims to replace traditional cookstoves with an efficient woodfuel cookstove technology. All the traditional three stones cookstoves for domestic use used by the different wives (if polygamous) of one household will be replaced with project stoves with an efficiency of at least 20%[15]. The sizes 2, 3, 4, 5, 6, 7, 8, 10, 12 and 15 will be taken in this VPA.
2	Baseline	The baseline fuel is only firewood and the baseline stove is a three stone fire, or a conventional device	The baseline scenario is non-renewable firewood consumption to meet thermal energy requirement for household cooking. The Baseline Field Performance

[14] See document 'Fiche technique de construction de F3PA_tiipaalga'

[15] Rapport sur les tests de performances énergétiques des Foyers trois pierres améliorés (F3PA) de l'association Tiipaalga, Laboratoire Biomasse Energie et Biocarburnat de 2IE, Ouagadougou, July 2015 (see document '*tiipaalga_Rapport de tests de performance énergétiques_F3PA_24_07_2015_VF.pdf*')

		without a grate or a chimney i.e. with no improved combustion air supply or flue gas ventilation, as per Gold Standard Simplified Methodology for Efficient Cookstoves.	Test[16] realized in the project boundary shows that all households use firewood as the main baseline fuel. It appears that no other type of fuel is used for domestic cooking. From the report of the Baseline survey[17], it appears that 100% of the households use traditional/three stone stoves for domestic cooking.
3	Boundary and location of the VPA	The project activity is located within Burkina Faso	The VPA is located in the rural municipalities of Arbole, Bagare, Bokin, Samba, Pilimpikou, Kirsi and Yako in the province of Passoré in the North Region of Burkina Faso.
4	Micro-scale limit for VPAs	The VPA will remain under the limit of 10,000 tonnes of CO_2e	The stoves installed in the VPA are expected to represent an annual CO_2 reduction of less than 10,000 tonnes of CO_2e.
5	Use of the baseline cookstove	The use of the baseline cookstove, as a backup or auxiliary technology, in parallel with the improved cookstove introduced by the project activity is permitted as longs as a mechanism is put into place to encourage the removal of the old cookstove and there is a definitive discontinuity of its use. The project documentation must provide a clear description of the approach chosen and the monitoring plan must provide a good understanding of the extent to which the baseline technology is still in use after the introduction of the improved technology (whether the existing baseline cookstove is not surrendered at the time of the introduction of the improved technology, or whether a new baseline cookstove is acquired and put to use by targeted end users during the project	In the VPA, awareness workshops with stove users on the multiple advantages of the efficient F3PA cookstoves will be conducted before the construction of the new project cookstoves. In most cases the baseline stove will automatically be removed as the stones will be integrated in the new efficient F3PA cookstoves and as in the kitchen the fixed project stove will be constructed on the place of the baseline stove. The usage of a baseline cookstove will be limited to exceptional events, like celebrations, in case the household do not dispose of an efficient F3PA cookstove with size 20 (for big cooking pot sizes). The monitoring plan describes how to measure the usage of the baseline technology during crediting period of the VPA through the to be monitored parameter $DF_{b,Stove,y}$.

[16]See document 'GS1340_VPA25-28_BFPT_20210430_Analysis'
[17]See document 'GS1340_VPA25-28_Baseline survey_20210430_Analysis'/Tab 'Report'

		crediting period). The success of the mechanism put into place must therefore be monitored, and the approach must be adjusted if proven unsuccessful. If the baseline cookstove remains in used in parallel with the project cookstove, corresponding emissions must be accounted for as part of the project emissions.	
6	Avoiding Double Counting of Emissions Reductions	Each VPA will ensure double counting of emission reductions is avoided through a unique numbering or identification system for the disseminated stoves	When a household is included in the VPA, all traditional three stones cookstoves for domestic use used by the different wives (if polygamous) of one household will be replaced with project cookstoves. The project activity has set up a monitoring system whereby each wife of the household included in the project activity will be registered in a database on Akvo software[18] with a unique serial number referring to this micro scale – VPA: GS1340-VPA-25-xxxx/y where xxxx is the number of the household (1 to 9999) and y is the number of the wife in the household.
7	VER ownership	End users receiving efficient woodstove under the specific VPA contractually cede their rights to claim and own emission reductions under the Gold Standard to PP.	Each wife of one households included in the project will sign a waiver[19] for the transfer of the credit to PP.
8	The Micro-scale-VPA-DD has been reviewed by the CME and submitted to the GS for inclusion into the PoA ;	The VPA implementer shall submit a Micro-scale-VPA-DD to the CME for each Micro-scale-VPA and with all underlying evidence.	See document 'GS1340_Technical_review_of_VPA_inclusion_VPA-25-28_signed'
9	Non-Diversion of ODA	There will be no diversion of ODA for any of the proposed VPA's.	A declaration confirming that there is no public funding for this VPA is available for the project activity.[20]

[18] www.akvo.org

[19] The final carbon waiver for this project is not yet finished as it is still in the design phase (by an graphical illustrator). The text that will be included can already be consulted in the document '*Contrat dessin F3PA_Projet Passoré_FR* '. The final version will be submitted at first performance certification as proof start of project..

[20] See document : '501_V1.0_AR_GHGs_ODA_GS1340_VPA-25_GS11070_signed'

10	Avoiding Double Counting of Programme Activities	Each VPA will show that it is exclusive to the PoA and not registered as another project activity or VPA under another PoA.	This VPA is neither registered as a project activity with GS or any other standard or as a VPA of another PoA. The relevant online carbon registries (CDM, GS, VCS) have been checked to confirm that there are no other domestic ICS projects in the Passoré province, hence no risk of double counting in this province.
11	Local stakeholder consultation	Each VPA will conduct a local stakeholder consultation (LSC) in order to gain feedback from stakeholders representing the specific project areas. A single LSC meeting can be organised for several micro-scale project activities if approved by The Gold Standard Foundation.	The single Local Stakeholder Consultation meeting has been conducted in Yako on Thursday 25th February 2021. The LSC report[21] includes a description of how local stakeholders were invited and a summary of the first comments received after sending the invitations. The arguments for organizing a single LSC for the 4 VPAs are (i) the 7 municipalities included in the 4 grouped VPA's are very close to each other (furthest distance of 110 km); (ii) the intention is to implement the 4 VPAs within the timeframe of 3 years; (iii) all 4 VPAs will apply the same distribution/implementation mechanism; (iv) the technology used in all 4 VPA's will be the same; and (v) the 7 municipalities where the VPA's will take place, are characterized by very similar socio economic situations.
12	Start date of the VPA	The VPA shall not begin before the date of registration of the PoA	The start date of the project is considered as the day of installation of the first F3PA in the project area. The start date of the VPA 25 is 10/03/2021[22].
13	Environmental impact assessment	Each VPA will conduct an environmental impact assessment or provide by the Ministry of Environment a letter of exemption for the environmental impact assessment.	The Letter of exemption[23] for the environmental impact assessment has been provided by the Ministry in charge of the environment of Burkina Faso.

[21] See LSC report 'T-PreReview_V1.1- SCR - Solidagro_GS1340_VPA25-28'

[22] See carbon waiver : 'Carbon waiver - VPA 25 - HH 35_with comments'.

[23] See document 'Lettre Exemption EIE - Min. Env. B'

14	Target groups and distribution mechanisms	Target groups eligible under this PoA are rural or urban households. The mechanisms for distribution of efficient cookstoves under this PoA are direct distribution/installation, delivery, community sales events, direct sales or sales through commercial/retail outlets.	As described in the VPA-DD the target groups are rural households. The distribution mechanism of the efficient F3PA cookstoves is direct distribution and installation.
15	Prior consideration of the carbon revenues in case of retroactive VPA	In case of retroactive VPA, it shall be demonstrated that carbon finance was a decisive factor to implement the VPA.	N/A Project cycle is regular.

The General Eligibility Criteria as per "Gold Standard For The Global Goals Principles & Requirements" is described below:

(a) **Types of Project:** The project aims to replace traditional cookstoves with an efficient woodfuel cookstove technology. Please refer to point 1 of table "eligibility criteria as defined in approved PoA"

(b) **Location of Project:** The location of the project is in line with the eligibility criteria defined as per approved PoA. Please refer to point 3 of table "eligibility criteria as defined in approved PoA"

(c) **Project Area, Project Boundary and Scale:** The project activity is located within Burkina Faso. The implementation will cover the municipalities of Arbole, Bagare, Bokin, Samba, Pilimpikou, Kirsi and Yako in the province of Passoré in Burkina Faso. The scale of the project will be within the threshold limit of micro scale project i.e. the VPA is expected to represent an annual CO2 reduction of less than 10,000 tonnes of CO2e. This VPA is neither registered as a project activity with GS or any other standard or as a VPA of another PoA.

(d) Please refer to point 3, 4 and 10 of table "eligibility criteria as defined in approved PoA "**Host Country Requirements:** The Projects implementation is in compliance with Burkina Faso's legal, environmental, ecological and social regulation. For reference, the Letter of exemption[24] for the environmental impact assessment has been provided by the Ministry in charge of the environment of Burkina Faso.

(e) **Contact Details:** The name and contact details of all the project participants is included in the PDD under Appendix I

(f) **Legal Ownership:** End users receiving efficient woodstove under the specific VPA contractually cede their rights to claim and own emission reductions under the

[24] See document 'Lettre Exemption EIE - Min. Env. BF'

Gold Standard to PP, which are Solidagro and CO2logic. Please refer to point 7 of table "eligibility criteria as defined in approved PoA"

(g) **Other Rights:** The project is voluntary initiated for the development of community. The target groups are rural households and there are no legal rights contested within the project.

(h) **Official Development Assistance (ODA) Declaration:** A declaration confirming that there is no public funding for this VPA is available for the project activity. Please refer to point 9 of table "eligibility criteria as defined in approved PoA"

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

The project owners, CO2logic and Solidagro, have full and uncontested legal ownership of the carbon credits that are generated under the Gold Standard Certification. Each end user included into the project will have to sign a waiver[25] for the transfer of credit to project developers.

A.2. Location of project

Country: Burkina Faso

Province: Passoré

Municipalities:

- Arbolé
- Bagare
- Bokin
- Samba
- Pilimpikou
- Kirsi
- Yako

Geographical location

[25] See document : 'Carbon waiver - VPA 25 - HH 35_with comments'.

Province	Municipality	Latitude	Longitude
Passoré	Arbolé	12.8433	-2.0385
Passoré	Bagare	13.3022	-2.3582
Passoré	Bokin	13.0027	-1.7928
Passoré	Samba	12.6682	-2.3811
Passoré	Pilimpikou	12.7017	-2.2703
Passoré	Kirsi	13.0370	-1.9375
Passoré	Yako	12.9515	-2.2550

Map of the project intervention area

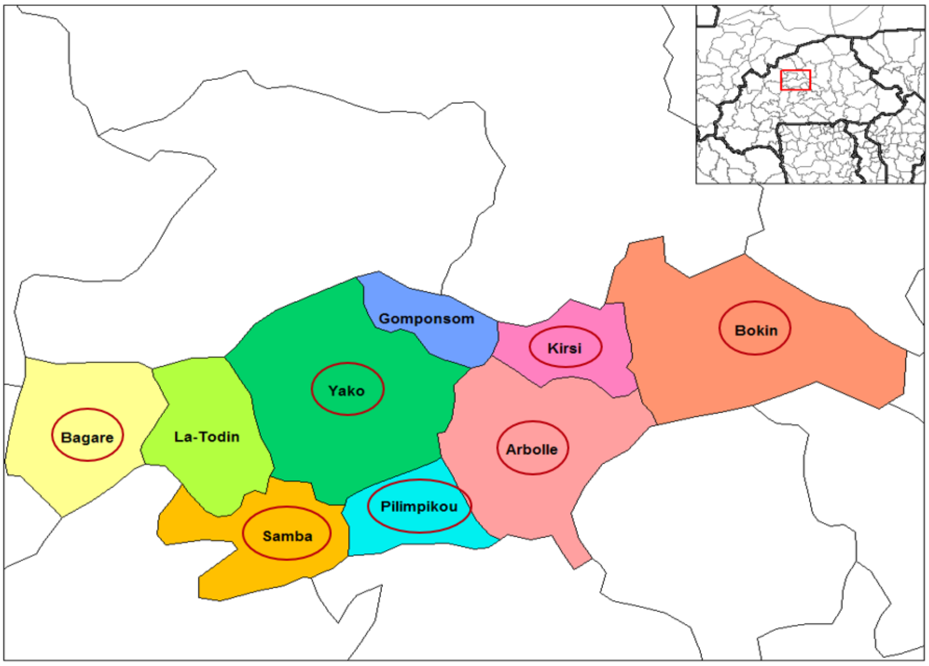

Figure 2 Map of the project intervention zone

A.3. Technologies and/or measures

The micro-scale VPA involves assisting stove users in the rural provinces of Passoré in the North region of Burkina Faso to gain the skills required to build, use and maintain their own efficient F3PA cookstoves.

Description of the technology

The F3PA efficient cookstove is a technology initially developed by Association tiipaalga who agreed to transfer this technology to project participants. F3PA cookstoves are made from mud mixtures with inexpensive locally sourced materials: manure, straw, water, and clay (or soil coming from termite mound). These project stoves consist of a thick circular wall of mud surrounding and shielding a pot and have one entrance for the woodfuel directly beneath the pot. The pot itself is supported by three stones.

Figure 3- F3PA efficient cookstoves

The F3PA efficient cookstoves are single pot stoves, which means that every cooking pot size has its specific size of cookstove. The sizes of the cooking pots and so the cookstoves used in this VPA are 2, 3, 4, 5, 6, 7, 8, 10, 12 and 15 due to its frequency of utilization. The diameter of cooking pots of size 2 and 15 varies between 22 cm and 42cm, which will influence the external diameter of project cookstoves. The average measures of the project cookstoves can be found in the table below. The distance between the cooking pot and the floor of the cookstove determines the size of the combustion chamber and thus influences the thermal efficiency of the project stove. If the combustion chamber is too small, the cooking pot will be surrounded with lots of flames and the consumption of wood will be higher. If the combustion chamber is too large, too much energy will be lost. The standard measure for the distance between the cooking pot and the floor of the cookstove of the F3PA promoted by project participants will vary between 15 cm and 20 cm depending on the size of the cooking pot. The distance between the cooking pot and the wall of cookstove allows having good ventilation and functions as a stack. The standard measure for the distance between the cooking pot and the wall of the F3PA cookstove is about 3 cm. The women measure this distance with the thickness of their hand. The shape and measure of the wood entrance of the cookstove are also important features for the performance of the F3PA project cookstoves. If the entrance is too large, the energy loss will be high and the cookstove becomes fragile. If the entrance is too small, the supply of wood is hampered which could cause cracks and the air ventilation is not assured. The circular shape of the entrance diminishes cracks around the entrance and reinforces the resistance of the wall of the cookstove. The standard measure of the wood entrance of the F3PA will not exceed half the height of the cookstove.

Size of cooking pot	External diameter of pot (cm)	Distance between cooking pot and wall of cookstove (cm)	Distance between cooking pot and floor of cookstove (cm)	Height of wood entrance (cm)	Diameter of wood entrance (cm)	Height of F3PA cookstove (cm)	Diameter F3PA cookstove (cm)
2	22	3	15	12	12	34	34
3	25	3	15	13	13	37	37
4	27	3	15	18	18	42	42
5	29	3	15	18	19	45	45
6	31	3	17	19	19	47	47
7	32	3	17	20	20	50	48
8	34	3	17	22	20	53	52
10	37	3	20	23	22	55	54
12	40	3	20	25	23	57	60
15	42	3	20	27	25	64	65

Table 1 Dimensions of F3PA promoted in the project

All F3PA stoves of different sizes have a specified efficiency of at least 20%[26]. Based on other project experiences of tiipaalga, the efficient F3PA cookstove has a life span of at least five years, as the women take an active role to undertake any repairs that may be required in the future. Training and monitoring are provided by the leader women to the women to teach how to maintain and repair their own mud made 3 stones efficient woodstoves. The instructors of APEDR, trained by Association tiipaalga, will be available to assist the leader women and F3PA owners how to maintain and repair their stoves. The use of virtually unlimited, locally available, natural materials to construct the stove allows repairs to be conducted as required at little cost.

Distribution mechanism

[26] Rapport sur les tests de performances énergétiques des Foyers Trois Pierres Améliorés (F3PA) de l'Association Tiipaalga, Laboratoire Biomasse Energie et Biocarburant de 2IE, Ouagadougou, July 2015 (see document « tiipaalga Rapport de tests de performances énergétiques_F3PA_24_07_2015_VF.pdf »)

A learning pyramid will be set up in order to have an efficient distribution mechanism of the efficient F3PA cookstoves across the households within the project boundary. This system will ensure that the required number of local women is trained to build, use and maintain their efficient F3PA cookstove. At the top of the pyramid one will find a project coordinator with University Degree profiles. He will manage a certain number of female animators and ensure they have the required level of training and skills to pass down their F3PA construction knowhow to the next level of the pyramid. The animators have the required on the ground knowledge and understanding of the context in the intervention zone. Each animator will manage and train a certain number of leader women (Monitrice Endogène) in the various villages every year. These leader women are selected by the other women in the village and represent the more influential and knowledgeable members of the community who have the capacity to pass on their acquired understanding of F3PA construction skills. These leader women will thus educate and inform a specific quantity of women in the villages to in turn build, use and maintain their own efficient F3PA cookstoves. The efficient F3PA cookstoves have a strict construction protocol[27] with diagrammatic presentation material[28] to support the APEDR animator and the leader women on the ground and ensure that project stoves are systematically built in a similar manner.
With adequate maintenance by the stove users it is expected that the efficient F3PA cookstoves will last for more than five years. Leader women are trained to evaluate the condition of the stoves. In case that the stove is not in 'good condition', the efficient F3PA cookstove will be reconstructed under supervision of the leader women. Quality control will be guaranteed by the animators.

A.4. Scale of the project

The project is a micro-scale project. The emission reductions made possible by the use of the stoves are expected to represent 10,000 tCO2e annually.

A.5. Funding sources of project

The necessary finance for this project will be from project developers own funds (Solidagro and CO2logic). Carbon finance associated to the project will allow to monitor and replace all the project stoves at the end of their lifespan. Part of those revenues will also be used to extend the project to the whole project intervention area.

[27] See document « Fiche technique de construction de F3PA_tiipaalga »
[28] See document « tiipaalga -Boite à images » and document « Guide d'animation_tiipaalga_vf »

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

The Gold Standard Simplified Methodology for Efficient Cookstoves Version 1.0

B.2. Applicability of methodology (ies)

Methodology applicability requirement	Justification related to project activity
This methodology is applicable to the micro-scale programmes and micro-scale activities introducing new wood fired cookstoves that reduce use of non-renewable fire wood or switch from non-renewable to renewable fire wood to meet thermal energy requirement for household cooking.	The project is a VPA of a micro-scale programme and introduces new wood fired cookstoves that reduce use of non-renewable firewood. The targeted end-users are households which use a three stone open fire and are lacking access to improved cooking technology in the absence of this project activity. The introduced improved cookstove technology is the F3PA efficient cookstove, which has successfully been introduced by Association tiipaalga in other provinces of Burkina Faso, i.e. Bam, Lorum, Kourweogo and Nahouri.
A project proponent implements the micro-scale activity or micro-scale programme of activities. The individual households and institutions do not act as project proponents.	Solidagro, APEDR and CO2logic will implement the project with the technical support of Association tiipaalga. No institutions will be involved in the project. The individual households do not act as project proponent.
This methodology is applicable, *i. If the baseline fuel is only fire wood;* *ii. If the baseline stove is a three stone fire, or a conventional device without a grate or a chimney i.e., with no improved combustion air supply or flue gas ventilation;* *iii. If the project stove is single pot or multi pot portable or in-situ cook*	i. The Baseline Field Performance Test[29] realized in the project boundary shows that all households use firewood as the main baseline fuel. It appears that no other type of fuel is used for domestic cooking. ii. From the report of the Baseline survey[30], it appears that 100% of the households use

[29]See document 'GS1340_VPA25-28_BFPT_20210430_Analysis'
[30]See document 'GS1340_VPA25-28_Baseline survey_20210430_Analysis'/Tab 'Report'

stoves with specified efficiency of at least 20%.	traditional/three stone stoves for domestic cooking. iii. The project stove is the F3PA efficient cookstove, which is a single pot and in-situ cookstove. The F3PA efficient cookstove may have different sizes according to the used cooking pot, i.e. sizes 2, 3, 4, 5, 6, 7, 8, 10, 12 and 15. All F3PA stoves of different sizes have a specified efficiency of at least 23.4%[31].
The project boundary can be clearly identified, and the cookstoves counted in the proposed project activity are not included in another voluntary market or CDM project activity (i.e. no double counting takes place). The project proponent must have a mechanism in place together with appropriate mitigation measures to prevent double counting.	The project activity is implemented in the province of Passoré in Burkina Faso. Details about the project location are provided in section A.2 of this document. The unique identification including GPS coordinates of each household where all traditional three stones cookstoves for domestic use used by all the different wives within the household (if polygamous) have been replaced with project stoves, will be recorded in AKVO[32] software. Moreover, the relevant online carbon registries (CDM, GS, VCS) have been checked to confirm that there are no other domestic ICS projects in the Passoré province, hence no risk of double counting in this province.
The project proponent must clearly communicate that the entity is claiming ownership rights and selling of the emission reductions resulting from the project activity. This must be communicated to the efficient cookstoves producers, retailers and end users by contract or clear written assertions in the transaction paperwork.	The ownership and selling rights of the emission reductions resulting from the project activity are clearly defined, defining Solidagro and CO2logic as the entities claiming ownership rights of and selling the emission reductions resulting from the project activity. Each stove user will sign a carbon waiver[33] confirming the transfer of

[31] Rapport sur les tests de performances energétiques des Foyers Trois Pierres Améliorés (F3PA) de l'Association Tiipaalga, Laboratoire Biomasse Energie et Biocarburant de 2IE, Ouagadougou, July 2015 (see document 'tiipaalga Rapport de tests de performances énergétiques_F3PA_24_07_2015_VF.pdf ')

[32] www.akvo.org

[33] See document: 'Carbon waiver - VPA 25 - HH 35_with comments'.

	carbon credit ownership from stove users to Solidagro and CO2logic.
The use of the baseline cookstove as a backup or auxiliary technology in parallel with the improved cookstove introduced by the project activity is permitted as long as a mechanism is put into place to encourage the removal of the old cookstove (e.g. discounted price for the improved cookstove) and the definitive discontinuity of its use. The project documentation must provide a clear description of the approach chosen and the monitoring plan must allow for a good understanding of the extent to which the baseline technology is still in use after the introduction of the improved technology, whether the existing baseline cookstove is not surrendered at the time of the introduction of the improved technology, or whether a new baseline cookstove is acquired and put to use by targeted end users during the project crediting period. The success of the mechanism put into place must therefore be monitored, and the approach must be adjusted if proven unsuccessful. If the baseline cookstove remains in use in parallel with the project cookstove, corresponding emissions must of course be accounted for as part of the project emissions.	During the implementation of the project activity awareness workshops with stove users will be conducted on the multiple advantages of the efficient F3PA cookstoves before the construction of the new project cookstoves. In most cases the baseline stove will automatically be removed as the stones will be integrated in the new efficient F3PA cookstoves and as in the kitchen the fixed project stove will be constructed on the place of the baseline stove. The usage of a baseline cookstove will be limited to exceptional events, like celebrations, in case the household do not dispose of an efficient F3PA cookstove with size 20 (for big cooking pot sizes). The monitoring plan describes how to measure the usage of the baseline technology during crediting period of the project activity.

B.3. Project boundary

According to the methodology, the *project boundary* is the physical, geographical site of baseline and project cookstoves and fuel collection area. In the scope of this project, it is the physical location of the households using the efficient F3PA cookstove after the implementation of the project activity in the province of Passoré (municipalities of Arbole, Bagare, Bokin, Samba, Pilimpikou, Kirsi and Yako) as well as the fuel collection areas which are the locations where the households source their wood for domestic cooking. The target area are the locations of the households using the efficient F3PA cookstoves after implementation of the project activity, where the usage of the open three stone fire as baseline cookstove is prevalent and uniform.

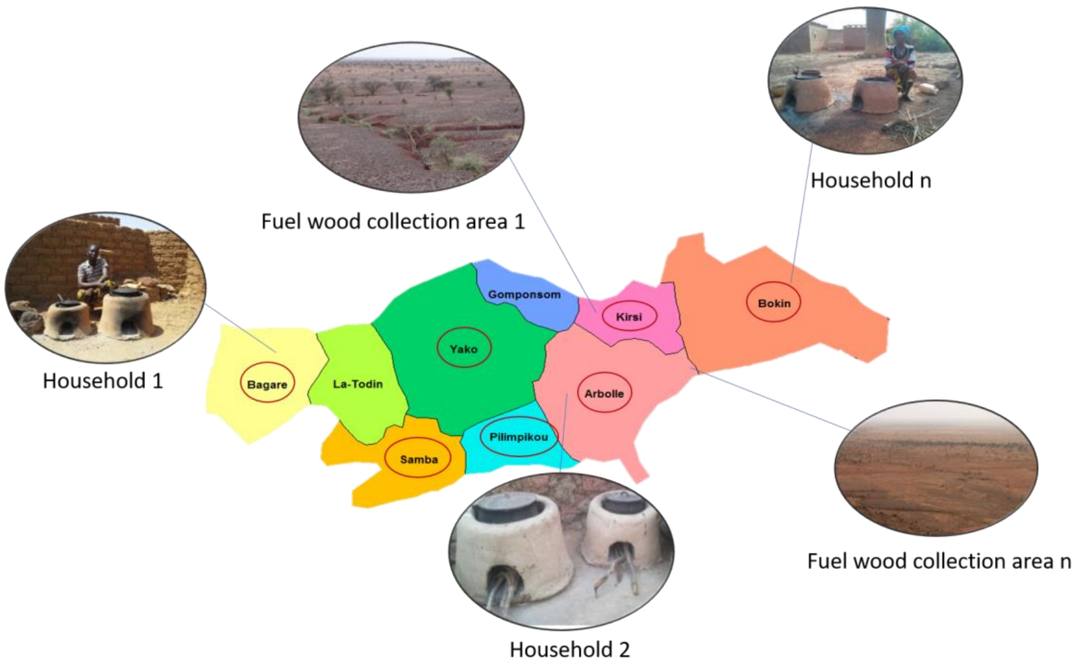

Figure 4 Diagram of project boundary and target area of the project activity in the 7 municipalities in province of Passoré

Source		GHGs	Included?	Justification/Explanation
Baseline scenario	Heat delivery	CO_2	Yes	Important source of emissions as CO2 is emitted any time biomass or fossil fuels are burned.
		CH_4	Yes	Important source of emissions released during partial or incomplete combustion of biomass during domestic cooking.
		N_2O	Yes	Important source of emissions released during partial or incomplete combustion of biomass during domestic cooking.
Project scenario	Heat delivery	CO_2	Yes	Important source of emissions as CO2 is emitted any time biomass or fossil fuels are burned.
		CH_4	Yes	Important source of emissions released during partial or incomplete combustion of biomass during domestic cooking.
		N_2O	Yes	Important source of emissions released during partial or incomplete combustion of biomass during domestic cooking

B.4. Establishment and description of baseline scenario

In accordance with the methodology, the baseline scenario is non-renewable firewood consumption to meet thermal energy requirement for household cooking. A Baseline Survey and Baseline Field Performance Test have been conducted between 03/03/2021 and 13/03/2021 in 8 randomly selected villages from the 35 villages located in the project boundary, i.e. the municipalities municipalities of Arbolé, Bagare, Bokin, Kirsi, Pilimpikou, Samba and Yako in the Province of Passoré.

Within those 8 villages, 122 households were randomly selected in order to meet the minimum 100 sample size requirement for the Baseline Survey and the 95/10 confidence precision requirement for the mean value for baseline fuel consumption in the Baseline Field Performance Test across the VPA's. The sampling approach for the selection of the 8 villages is described in document 'BS BFCT Sampling_Solidagro_GS1340 VPA 25-28_v2.0'. From 27/02/2020 till 04/03/2020, the 4 surveyors received a training for the conduction of the Baseline Field Performance Test according to the requirements of the methodology. The document 'BFPTP_v.1.0_EN and FR_Solidagro_VPA 25-28_clean' describes the protocol for the realization of the Baseline Field Performance Tests. A complete report of the training can be found in document: 'Report training BFPT_Solidagro_with comments'.

The Baseline Field Performance Test[34] realized in the project boundary, shows that 100% of the used energy for domestic cooking comes from firewood. The Baseline Survey[35] report shows that 100% of the surveyed households use the traditional/three stone fire as baseline stove and 96% have never cooked on an improved cookstove. The wide majority of the surveyed households are currently not satisfied with their stoves and estimate that the use of firewood causes environmental degradation problems. Overall, the results of the survey indicated an average consumption of 3.53 tonnes of wood per household per year for domestic cooking. Information regarding the surveyed sample including pictures of the baseline stoves is found in document 'GS1340_VPA25-28_Baseline Survey_Pictures of baseline stoves'.

The concept of suppressed demand is not applied.

B.5. Demonstration of additionality

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version	The proposed project follows the "Gold Standard for the Global Goals Community Services Activity Requirements, Version 1.2, October 2019":as

[34] See document : 'GS1340_VPA25-28_BFPT_20210430_Analysis'
[35] See document : 'GS1340_VPA25-28_Baseline survey_20210430_Analysis'

number and the specific paragraph, if applicable).	- the project leads to climate change mitigation and/or adaption by providing access to thermal energy services at household level; - the type of project is end-use energy efficiency, i.e. efficient cooking; - the project provides a clear description, of the ownership of the Products that are generated under the Gold Standard Certification Projects that meet any one of the criteria defined below (and meet the eligibility requirements) shall be deemed additional (see section 4.1.9 of requirements): (b) The project is located in a Least Developed Country (LDC), Small Island Developing States (SIDS) or Land Locked Developing Country (LLDC)
Describe how the proposed project meets the criteria for deemed additionality.	The project activity is located in Burkina Faso, which is an LDC (https://unohrlls.org/about-ldcs/), therefore the activity is automatically deemed additional.

The GS has published Community Services Activity Requirements; the proposed project falls under the automatic eligibility list of projects. Project activity falls under End-use Energy efficiency which reduces energy requirement as compared to baseline scenario without affecting the level and quality of services. The project will not adversely affect men and women in marginalized or vulnerable communities. Implementation of the project will contribute towards preservation of common resources in form of “firewood”. Household duties related to firewood collection, cooking and cleaning utensils remain with women. The project therefore tends to decrease burden on women and will not result in social isolation of men.
The project leads to safe working condition and improvement in health as it will reduce firewood consumption as fuel with improved cook stove which is efficient and safe. The usage of improved cook stove will improve the health of family especially mother and children and support in improving the life by reducing the cooking and fuelwood collection time.

B.5.1. Prior Consideration

N/A

B.5.2. Ongoing Financial Need

N/A

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact Indicator (Proposed or SDG Indicator)
1. No poverty	1.1 By 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than $1.25 a day	Indicator 1: Average household savings i.e., decrease in expenditure for woodfuel for domestic cooking Indicator 2: Total estimated amount saved by stove users on wood fuel purchase
3. Good Health and well being	3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination	Proportion of households perceiving less often smoke levels, incidence of coughing, incidence of respiratory illness, incidence of itchy eyes
4. Quality Education	4.3. By 2030, ensure equal access for all women and men to affordable and quality technical, vocational and tertiary education, including university.	Number of workshops carried out for women
4. Quality Education	4.4 By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship	Number of employee provided skill development training

5. Gender Equality	5.4 Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate	Average time saving associated with cooking time and fuel collection
5. Gender Equality	5.5 Ensure women's full and effective participation and equal opportunities for leadership at all levels of decision-making in political, economic and public life	Number of women serving in managerial/ leadership /ownership role
7. Affordable clean energy	7.1 By 2030, ensure universal access to affordable, reliable and modern energy services	Number of Households
13. Climate Action	13.2 Integrate climate change measures into national policies, strategies and planning	Amount of GHGs emissions avoided.
15. Life on land	15.1 By 2020, ensure the conservation, restoration and sustainable use of terrestrial and inland freshwater ecosystems and their services, in particular forests, wetlands, mountains and drylands, in line with obligations under international agreements	Total non-renewable wood fuel saved

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

a) <u>SDG1, No poverty</u>

Monitoring indicator:

Average household savings i.e., decrease in expenditure on wood fuel purchase = estimated amount in FCFA or € saved in average by the stove users for the purchase of wood fuel during the monitoring period

Total estimated amount saved by stove users on wood fuel purchase for the group of VPA's= total estimated amount in FCFA or € saved by the stove users of the group of VPA's on wood fuel purchases during the monitoring period

Baseline: based on the Baseline Survey report[36], it appears that 21% of the households purchase wood fuel for domestic cooking. The average yearly spent amount is estimated to be 14,442 FCFA or 22 € per household who purchases wood. Considering 2,800 household per VPA, the total yearly amount for the 4 VPAs is estimated to be 45,192 € on average.

Usage of saved money by women:

School fees$_p$ = (Number of women using their saved money for the payment of school fees) / (Number of women considering they save money thanks to the F3PA efficient cookstoves)

Purchase of medical drugs$_p$ = (Number of women using their saved money for the purchase of medical drugs) / (Number of women considering they save money thanks to the F3PA efficient cookstoves)

Purchase of food$_p$ = (Number of women using their saved money for the purchase of food) / (Number of women considering they save money thanks to the F3PA efficient cookstoves)

Investment for field crops$_p$ = (Number of women using their saved money to invest in field crops) / (Number of women considering they save money thanks to the F3PA efficient cookstoves)

Purchase of equipments$_p$ = (Number of women using their saved money to purchase equipments like mobile, bicycle, …) / (Number of women considering they save money thanks to the F3PA efficient cookstoves)

Income generating activities$_p$ = (Number of women using their saved money for income generating activities) / (Number of women considering they save money thanks to the F3PA efficient cookstoves)

Savings$_p$ = (Number of women using their saved money for their savings) / (Number of women considering they save money thanks to the F3PA efficient cookstoves)

b) SDG 3, Good health and well being

Monitoring indicators

Smoke level reduction = (Number of stove users perceiving less smoke since the implementation of F3PA efficient cookstoves) / (Number of respondents)

[36] See document : 'GS1340_VPA25-28_Baseline survey_20210430_Analysis'

Baseline: during monitoring survey the smoke level perceived by the stove user will be compared between the baseline scenario before the implementation of the F3PA efficient cookstove and the project scenario since the implementation of the F3PA efficient cookstove.

Incidence of coughing reduction = (Number of stove users perceiving less incidence of coughing since the implementation of F3PA efficient cookstoves) / (Number of respondents)

Baseline: during monitoring survey the incidence of coughing perceived by the stove user will be compared between the baseline scenario before the implementation of the F3PA efficient cookstove and the project scenario since the implementation of the F3PA efficient cookstove.

Incidence of respiratory illness reduction = (Number of stove users perceiving less incidence of respiratory illnesses since the implementation of F3PA efficient cookstoves) / (Number of respondents)

Baseline: during monitoring survey the incidence of respiratory illness will be compared between the baseline scenario before the implementation of the F3PA efficient cookstove and the project scenario since the implementation of the F3PA efficient cookstove.

Incidence of itchy eyes reduction = (Number of stove users perceiving less incidence of itchy eyes since the implementation of F3PA efficient cookstoves) / (Number of respondents)

Baseline: during monitoring survey the incidence of itchy eyes perceived by stove users will be compared between the baseline scenario before the implementation of the F3PA efficient cookstove and the project scenario since the implementation of the F3PA efficient cookstove.

c) SDG 4, Quality education

Monitoring indicator

Number of workshops carried out for women for the group of VPA's = Number of workshops for the group of VPA's carried out for women during the monitoring period

Baseline: no workshops have been organized for women in the project boundary

Number of employee provided skill development training = Number of employee provided skill development training involved in the programme during the monitoring period

Baseline: no training initiatives have been organized for the staff members assigned to the project

d) SDG 5, Gender equality

Monitoring indicator

Average time saving associated with cooking time and fuel collection = Average time saving (in hours) on wood fuel collection per woman per week during the monitoring period

Baseline: based on the Baseline Survey report[37], it appears that on average women spend 7.2 hours per week on wood fuel collection

Activities carried out by women during saved time:

i. **Domestic tasks$_p$** = (Number of women using their saved time to do domestic tasks) / (Number of women considering they save time thanks to the F3PA efficient cookstoves)
ii. **Income generating activities$_p$** = (Number of women using their saved time to do income generating activities) / (Number of women considering they save time thanks to the F3PA efficient cookstoves)
iii. Field labour$_p$ = (Number of women using their saved time to do field labour) / (Number of women considering they save time thanks to the F3PA efficient cookstoves)
iv. Gardening$_p$ = (Number of women using their saved time to do gardening) / (Number of women considering they save time thanks to the F3PA efficient cookstoves)
v. **Participation to a literacy program$_p$** = (Number of women using their saved time to participate to a literacy program) / (Number of women considering they save time thanks to the F3PA efficient cookstoves)
vi. Community work$_p$ = (Number of women using their saved time to do community work) / (Number of women considering they save time thanks to the F3PA efficient cookstoves)
vii. **Religious activities$_p$** = (Number of women using their saved time to participate to religious activities) / (Number of women considering they save time thanks to the F3PA efficient cookstoves)

[37] See document: 'GS1340_VPA25-28_Baseline survey_20210430_Analysis'

Number of women serving in managerial/ leadership /ownership role = Number of leader Women or 'Monitrices Endogènes' involved in the distribution mechanism of the F3PA project stove

e) SDG 7, Affordable clean energy

Monitoring indicator

Number of households = Number of households provided with efficient F3PA project stove used for domestic cooking

Baseline: no F3PA efficient cookstoves were disseminated yet in the baseline scenario.

f) SDG 13, Climate Action

The baseline scenario is considered by default fixed

In the project activity, cookstoves are installed at the start of the project activity or installed progressively, the baseline is considered by-default fixed until the end of the cookstoves (introduced in the project activity) useful life or the registered crediting period, whichever occurs earlier. If the project cookstove is replaced with a cookstove of similar efficiency prior to the end of the crediting period, the original baseline shall be applicable till the end of the replaced cookstoves useful life or the registered crediting period, whichever occurs earlier.

Only one project scenario is considered

The project scenario is the adoption of the efficient F3PA cookstove by end users in the target area of the municipalities of Arbole, Bagare, Bokin, Samba, Pilimpikou, Kirsi and Yako in the province of Passoré in the North Region, defined as the project boundary of the VPA.
Only one type of efficient cookstove will be installed, which is the efficient F3PA cookstove. Some households in the project boundary are composed of one husband with several wives. Each wife in a household will have at least two efficient F3PA cookstoves according to the local cooking requirement. All the traditional three stone cookstoves for domestic use used by the wives within the household in the VPA will be replaced by efficient F3PA cookstoves. Ten different sizes (2, 3, 4, 5, 6, 7, 8, 10, 12 and 15) are regularly used and will be taken account. The determination of quantity of firewood consumed in the baseline is at household level. For this reason, the number of households will be monitored instead of project cookstoves to determine the emissions reductions. The efficient F3PA cookstoves installed at a household can have different sizes according to the cooking habit within the household. If the efficiency of these F3PA cookstoves with different sizes differs, the lowest value will be taken as reference value for the efficiency of the cookstove being used in the project scenario to calculate the emission reductions.

Calculation of the emission reductions

The methodology directly provides equation for emission reductions (without separate baseline, projector leakage emission reduction equations). The emission reductions are calculated using the following equation:

$$ER_y = \sum_{0to1}^{xtoy} N_{P,y} * P_y * U_{P,y} * (f_{NRB,y} * EF_{b,fuel,CO2} + EF_{b,fuel,nonCO2}) * (1 - DF_{b,Stove,y})$$

Where

$N_{p,y}$	Number of households with project cookstoves of each age group operational in the year y;
P_y	Quantity of firewood that is saved in the year y (tones per household in year y);
$U_{p,y}$	Usage rate for project cookstoves in year y, based on adoption rate and drop off rate revealed by usage surveys (fraction);
$f_{NRB,y}$	Factional non-renewability status of wood fuel during year y
$EF_{b,fuel,CO2}$	CO2 emission factor of firewood that is substituted or reduced
$EF_{b,fuel,nonCO2}$	Non CO2 emission factor of firewood that is substituted or reduced
$DF_{b,stove,y}$	Usage of baseline cookstove during the year y (fraction) in project scenario;
X	y-1
Y	Year of the crediting period

<u>Determination of quantity of biomass saved (P_y):</u>

Quantity of firewood that is saved (P_y) is estimated using the following equation:

$$P_y = B_{b,y} * (1 - \frac{\eta_b}{\eta_{p,y}})$$

Where:

P_y	Quantity of firewood that is saved in the year y (tones per household in year y);
$B_{b,y}$	Quantity of firewood consumed in baseline scenario during year y (tones per household per year);
$\eta_{p,y}$	Efficiency of project cookstove in year y (fraction);

η_b	Efficiency of the baseline cookstove being replaced (fraction). A default value of 10% is used as the replaced cookstove is a three stone fire, or a conventional device without a grate or a chimney i.e. with no improved combustion air supply or flue gas ventilation;
y	Year of the crediting period;

Determination of quantity of fire wood consumed in the baseline ($B_{b,y}$):

The firewood consumed is the estimated average annual consumption of firewood per household (tones/year), which may be derived using option (d) of the methodology: baseline field performance test. Consumption is measured with a representative sample of end users under each defined baseline scenario using the baseline cookstove. A 95/10 confidence/precision must be met to use the mean value for baseline fuel consumption.

Determination of project cookstove efficiency ($\eta_{p,y}$ and η_p):

Efficiency of project cookstove in year y ($\eta_{p,y}$) is estimated as follows:

$$\eta_{p,y} = \eta_p * \left(DF_\eta\right)^{y-1} * 0.94$$

Where

$\eta_{p,y}$	Efficiency of project cookstove in year y (fraction)
η_p	Efficiency of project cookstove (fraction) determined at the start of the project activity
DF_η	Discount factor to account for efficiency loss of project cookstove per year of operation (fraction)
0.94	Adjustment factor to account for uncertainty related to project cookstove efficiency test

Calculation of leakage

As defined under The Gold Standard Simplified Methodology for Efficient Cookstoves, the net emission reductions (ERy) for a micro-scale program of activities (mPOA) need to be discounted by a factor of 0.95 to account for leakages related to non-renewable biomass saved by the project activity.

g) SDG 15, Life on land

Monitoring indicator

Total non-renewable wood fuel saved = Quantity of tonnes of wood saved due to the use of the F3PA cookstove during the monitoring period

Baseline: no F3PA efficient cookstoves were disseminated yet in the baseline scenario, hence the quantity of non-renewable wood saved is nil in baseline scenario.

B.6.2. Data and parameters fixed ex ante

SDG13

Data/parameter	**$EF_{b,fuel,CO2}$**
Unit	tCO_2/ton of firewood
Description	CO_2 emission factor arising from use of firewood in baseline scenario
Source of data	IPCC default values, table 1.4 of chapter 1 of Vol.2, 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value(s) applied	1.747 tCO_2/ton of firewood
Choice of data or Measurement methods and procedures	As defined under The Gold Standard Simplified Methodology for Efficient Cookstoves
Purpose of data	Calculation of emission reductions
Additional comment	N/A

Data/parameter	**$EF_{b,fuel,non\_CO2}$**
Unit	tCO_2/ton of firewood
Description	Non-CO_2 emission factor arising from use of firewood in baseline scenario
Source of data	IPCC Fifth Assessment Report
Value(s) applied	0.581 tCO_2/ton of firewood
Choice of data or Measurement methods and procedures	As per Rule Update "Applicability of Global Warming Potential for Gold Standard For The Global Goals Projects" published on 27/10/2020.

Purpose of data	Calculation of emission reductions
Additional comment	N/A

Data/parameter	η_b
Unit	Fraction
Description	Efficiency of the cookstove being used in the baseline scenario
Source of data	Gold Standard Simplified Methodology for Efficient Cookstoves
Value(s) applied	0.10
Choice of data or Measurement methods and procedures	As defined under The Gold Standard Simplified Methodology for Efficient Cookstoves
Purpose of data	Calculation of emission reductions
Additional comment	N/A

Data/parameter	η_p
Unit	Fraction
Description	Efficiency of the cookstove being used in the project scenario
Source of data	Determined following the Water Boiling Test Protocol

Value(s) applied	0.234[38]
Choice of data or Measurement methods and procedures	As defined under The Gold Standard Simplified Methodology for Efficient Cookstoves
Purpose of data	Calculation of emission reductions
Additional comment	For each wife of one household included in the VPA, at least two efficient cookstoves of the defined project sizes 2, 3, 4, 5, 6, 7, 8, 10, 12 and 15 will be installed according to the local cooking habits. Each size of project cookstove is tested according to the WBT protocol. To determine the project cookstove efficiency of one particular size, three sample runs have been carried out on one randomly selected project cookstove. The average of the three results is taken as the efficiency for the project cookstove of this particular size. The lowest value of project cookstove efficiency of the various sizes is taken as reference value for the efficiency of the cookstoves being used in the project scenario to calculate the emission reductions. The project cookstove efficiency in the year y $\eta_{p,y}$ will be determined using the discount factor DF_{η} to account for efficiency loss of project cookstove per year of operation (fraction).

Data/parameter	$f_{NRB,b,y}$
Unit	Fractional non-renewability
Description	Non-renewability status of wood fuel during year y

38 Ouagadougou, Laboratoire Biomasse Energie et Biocarburant, Rapport sur les tests de performances énergétiques des foyers trois pierres améliorés (F3PA) de l'Association tiipaalga (see document : 'tiipaalga_Rapport de tests de performance énergétiques_F3PA_24_07_2015_VF')

Source of data	Default NRB value provided by the CDM executive board and endorsed by the host country DNA (http://cdm.unfccc.int/DNA/fNRB/docs/burkina.pdf)
Value(s) applied	0.90
Choice of data or Measurement methods and procedures	As defined under The Gold Standard Simplified Methodology for Efficient Cookstoves
Purpose of data	Calculation of emission reductions
Additional comment	The project activity may choose to update the $f_{NRB,b,y}$ during the crediting period

Data/parameter	$\mathbf{B_{b,y}}$
Unit	Tonnes firewood per household per year
Description	Firewood consumption for cooking in the baseline
Source of data	Baseline Field Performance Test See document 'GS1340_VPA25-28_BFPT_20210430_Analysis'.
Value(s) applied	3.53
Choice of data or Measurement methods and procedures	Option d of Field Performance Test has been chosen to determine the firewood consumption for cooking in the baseline.
Purpose of data	Calculation of emission reductions
Additional comment	The baseline performance field test (BFT) measures real, observed performance of the baseline cookstove in the field. Consumption is measured with a representative sample of end users under each defined baseline scenario using the baseline cookstove. A 90/10 confidence/precision must be met to use the mean value for baseline fuel consumption. In case of sampling across VPA's 95/10 confidence precision must be met. A minimum sample size of 30 is recommended.

	A Baseline Survey and Baseline Field Performance Test have been conducted between 03/03/2021 and 13/03/2021 in 8 randomly selected villages from the 35 villages located in the project boundary, i.e. the municipalities municipalities of Arbolé, Bagare, Bokin, Kirsi, Pilimpikou, Samba and Yako in the Province of Passoré. Within those 8 villages, 122 households were randomly selected in order to meet the minimum 100 sample size requirement for the Baseline Survey and the 95/10 confidence precision requirement for the mean value for baseline fuel consumption in the Baseline Field Performance Test across the VPA's. The sampling approach for the selection of the 8 villages is described in document 'BS BFCT Sampling_Solidagro_GS1340 VPA 25-28_v2.0'. The protocol of the BFCT is found in document: 'BFPTP_v.1.0_EN and FR_Solidagro_VPA 25-28_clean'. Information regarding the surveyed sample including pictures of the baseline stoves is found in document ' GS1340_VPA25-28_Baseline Survey_Pictures of baseline stoves'.

B.6.3. Ex ante estimation of SDG Impact

Ex-ante calculations related to the outcome for SDG 13

Amount of GHGs emissions avoided = Detailed transparent ex-ante calculations of the outcome for SDG 13 (i.e. CO2e reductions) are available in a separate Excel Spreadsheet[39] based on the calculations presented in Section B.6.1.f.. From this, an average annual net emission reduction of 9,427 tCO2e is estimated.

Ex-ante calculations related to the outcome for SDG 1

Average household annual savings i.e., decrease in expenditure on wood fuel purchase = based on the baseline survey[40], it appears that 2,800

[39]See document : 'GS1340 - VPA-25 - ER – Calculation'
[40]See document : 'GS1340_VPA25-28_Baseline survey_20210430_Analysis'

households in one VPA spend annually in the baseline scenario 13,137 € for the purchase of wood (22€/household/year). It is estimated that thanks to the project the households will spend 30%[41] less money on the purchase of wood, i.e. savings of 7 € per household included in the project.

Total estimated amount saved by stove users on wood fuel purchase for the group of VPA's= based on the previous wood fuel purchase estimation (22€/household/year) and the number of stove users, the total estimated amount saved by stove users on wood fuel purchase for the group of VPA's can be derived assuming 2,800 households per VPA and a reduction of 30% in woodfuel purchase[42].

Ex-ante calculations related to the outcome for SDG 3

Smoke level reduction = it is estimated that more than 90% of the stove user respondents will perceive less smoke since the implementation of F3PA efficient cookstoves

Incidence of coughing reduction = it is estimated that more than 90% of the stove user respondents will perceive less incidence of coughing since the implementation of F3PA efficient cookstoves

Incidence of respiratory illness reduction = it is estimated that more than 90% of the stove user respondents will perceive less incidence of respiratory illnesses since the implementation of F3PA efficient cookstoves

Incidence of itchy eyes reduction = it is estimated that more than 90% of the stove user respondents will perceive less incidence of itchy eyes since the implementation of F3PA efficient cookstoves

Ex-ante calculations related to the outcome for SDG 4

Number of workshops carried out for women for the group of VPA's = it is estimated that at least about 80 workshops will have been organized for women for the group of 4 VPA's and per year or 72 per year in average for the whole period of the project. The 80 workshops should consist of:

- 15 general assembly workshops per year;
- 15 specific animation workshops per year;

[41] Estimation as the project foresees a wood fuel consumption reduction of 50%. To be conservative a reduction of 30% on expenditure on the purchase of wood has been taken.
[42] Estimation as the project foresees a wood fuel consumption reduction of 50%. To be conservative a reduction of 30% on expenditure on the purchase of wood has been taken.

- 20 training workshops for leader women per year;
- 15 awareness-raising workshops for women on the use and maintenance of improved stoves per year;
- 15 workshops per year to sensitize women on the protection of improved cookstoves against bad weather during the rains.

Number of employees provided skill development training for the group of VPA's = it is estimated that in the first year of the project 8 employees (3 women facilitators, one supervisor and 4 surveyors) will be provided skill development training and in the subsequent years 5 employees (5 surveyors).

Ex-ante calculations related to the outcome for SDG 5

Average time saving associated with cooking time and fuel collection = based on the baseline survey[43], it appears that a woman spends on average 7.2 hours per week on wood collection. It is estimated that thanks to the project, women will spend 30% less time on the collection of wood, i.e. time savings of 2.2 hours per woman per week.

Number of women serving in managerial/ leadership /ownership role = it is estimated that the number of leader women trained per VPA is 300 or for 1,080 women on average the group of VPA's .

Ex-ante calculations related to the outcome for SDG 7

Number of households = it is estimated that the number of households will be 2,800 households per VPA or on average 9,632 households per year for the group of VPA's.

Ex-ante calculations related to the outcome for SDG 15

Quantity of non-renewable wood saved due to the use of the F3PA cookstove = the calculation is based on the applied methodology:

$$= \sum_{0to1}^{xtoy} N_{P,y} * P_y * f_{NRB,y} * U_{P,y} * (1 - DF_{b,Stove,y})$$

Where

[43] See document : 'GS1340_VPA25-28_Baseline survey_20210430_Analysis'

$N_{p,y}$	Number of households with project cookstoves of each age group operational in the year y;
P_y	Quantity of firewood that is saved in the year y (tones per household in year y);
$f_{NRB,y}$	Factional non-renewability status of wood fuel during year y
$U_{p,y}$	Usage rate for project cookstoves in year y, based on adoption rate and drop off rate revealed by usage surveys (fraction);
$DF_{b,stove,y}$	Usage of baseline cookstove during the year y (fraction) in project scenario;
X	y-1

B.6.4. Summary of ex ante estimates of each SDG outcome[44]

Ex-ante estimation of SDG 13 Outcome

Year	Emission reduction (tCO2e/year)	Leakage	Net Emission Reduction (tCO2e/year)
Year 1	6,483	324	6,159
Year 2	10,714	536	10,000 (10,179)
Year 3	10,622	531	10,000 (10,091)
Year 4	10,530	526	10,000 (10,003)
Year 5	10,436	522	9,914
Year 6	10,341	517	9,824
Year 7	10,245	512	9,733
Year 8	10,149	507	9,641
Year 9	10,051	503	9,549
Year 10	9,953	498	9,455
Total	**99,524**	**4,976**	**94,275**

[44] Details regarding the calculations of SDG Impacts are available in the Document: 'GS1340 - VPA-25 - ER - Calculation'

Total number of crediting years	10		
Annual average over the crediting period	**9,952**	**498**	9,427

Ex-ante estimation of SDG 1 Outcome

Average household annual savings i.e., decrease in expenditure on wood fuel purchase

Year	Baseline estimate	Project estimate	Net benefit
Year 1	22 €	15 €	7 €
Year 2	22 €	15 €	7 €
Year 3	22 €	15 €	7 €
Year 4	22 €	15 €	7 €
Year 5	22 €	15 €	7 €
Year 6	22 €	15 €	7 €
Year 7	22 €	15 €	7 €
Year 8	22 €	15 €	7 €
Year 9	22 €	15 €	7 €
Year 10	22 €	15 €	7 €
Total	**22 €**	**15 €**	**7 €**
Total number of crediting years	10		
Annual average over the crediting period	**22 €**	**15 €**	**7 €**

Total estimated amount saved by stove users on wood fuel purchase for the group of VPA's[45]

Year	Baseline estimate	Project estimate	Net benefit
Year 1	€ 7,882	€ 5,518	€ 2,365
Year 2	€ 28,902	€ 20,231	€ 8,671
Year 3	€ 47,294	€ 33,106	€ 14,188
Year 4	€ 52,549	€ 36,784	€ 15,765
Year 5	€ 52,549	€ 36,784	€ 15,765
Year 6	€ 52,549	€ 36,784	€ 15,765
Year 7	€ 52,549	€ 36,784	€ 15,765
Year 8	€ 52,549	€ 36,784	€ 15,765
Year 9	€ 52,549	€ 36,784	€ 15,765
Year 10	€ 52,549	€ 36,784	€ 15,765
Total	**€ 451,921**	**€ 316,345**	**€ 135,576**
Total number of crediting years	10		
Annual average over the crediting period	**€ 45,192**	**€ 31,634**	**€ 13,558**

Ex-ante estimation of SDG 3 Outcome

Smoke level reduction

Year	Baseline estimate	Project estimate	Net benefit
Year 1	N/A	N/A	>90%
Year 2	N/A	N/A	>90%

[45] The following figures are evaluated based on an implementation plan of 1 VPA in Year 1, 2 in Year 2 and 1 VPA in year 3 and a reduction in wood purchase expenditures of 30%. It is estimated that VPAs are covered at 60% (or 1,440 households) in their first year of implementation.

Year 3	N/A	N/A	>90%
Year 4	N/A	N/A	>90%
Year 5	N/A	N/A	>90%
Year 6	N/A	N/A	>90%
Year 7	N/A	N/A	>90%
Year 8	N/A	N/A	>90%
Year 9	N/A	N/A	>90%
Year 10	N/A	N/A	>90%
Total	**N/A**	**N/A**	**>90%**
Total number of crediting years	10		
Annual average over the crediting period	N/A	N/A	>90%

Incidence of respiratory illness reduction

Year	Baseline estimate	Project estimate	Net benefit
Year 1	N/A	N/A	>90%
Year 2	N/A	N/A	>90%
Year 3	N/A	N/A	>90%
Year 4	N/A	N/A	>90%
Year 5	N/A	N/A	>90%
Year 6	N/A	N/A	>90%
Year 7	N/A	N/A	>90%
Year 8	N/A	N/A	>90%
Year 9	N/A	N/A	>90%
Year 10	N/A	N/A	>90%
Total	**N/A**	**N/A**	**>90%**
Total number of crediting years	10		

Annual average over the crediting period	N/A	N/A	>90%

Incidence of itchy eyes reduction

Year	Baseline estimate	Project estimate	Net benefit
Year 1	N/A	N/A	>90%
Year 2	N/A	N/A	>90%
Year 3	N/A	N/A	>90%
Year 4	N/A	N/A	>90%
Year 5	N/A	N/A	>90%
Year 6	N/A	N/A	>90%
Year 7	N/A	N/A	>90%
Year 8	N/A	N/A	>90%
Year 9	N/A	N/A	>90%
Year 10	N/A	N/A	>90%
Total	**N/A**	**N/A**	**>90%**
Total number of crediting years	10		
Annual average over the crediting period	N/A	N/A	>90%

Ex-ante estimation of SDG 4 Outcome

Number of workshops carried out for women per year for the group of VPA's[46]

[46] The following figures are evaluated based on an implementation plan of 1 VPA in Year 1, 2 in Year 2 and 1 VPA in Year 3.

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0	20	20
Year 2	0	60	60
Year 3	0	80	80
Year 4	0	80	80
Year 5	0	80	80
Year 6	0	80	80
Year 7	0	80	80
Year 8	0	80	80
Year 9	0	80	80
Year 10	0	80	80
Total	**0**	**720**	**720**
Total number of crediting years	10		
Annual average over the crediting period	0	72	72

Number of employees provided skill development training for the group of VPA's

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0	8	8
Year 2	0	5	5
Year 3	0	5	5
Year 4	0	5	5
Year 5	0	5	5
Year 6	0	5	5
Year 7	0	5	5
Year 8	0	5	5
Year 9	0	5	5
Year 10	0	5	5

Total	**0**	**53**	**53**
Total number of crediting years	10		
Annual average over the crediting period	0	5.3	5.3

Ex-ante estimation of SDG 5 Outcome

Average time saving associated with cooking time and fuel collection per woman per week

Year	Baseline estimate	Project estimate	Net benefit
Year 1	7.2 hours	5.0 hours	2.2 hours
Year 2	7.2 hours	5.0 hours	2.2 hours
Year 3	7.2 hours	5.0 hours	2.2 hours
Year 4	7.2 hours	5.0 hours	2.2 hours
Year 5	7.2 hours	5.0 hours	2.2 hours
Year 6	7.2 hours	5.0 hours	2.2 hours
Year 7	7.2 hours	5.0 hours	2.2 hours
Year 8	7.2 hours	5.0 hours	2.2 hours
Year 9	7.2 hours	5.0 hours	2.2 hours
Year 10	7.2 hours	5.0 hours	2.2 hours
Total	**N/A**	**N/A**	**N/A**
Total number of crediting years	10		
Annual average over the crediting period	**7.2 hours**	**5.0 hours**	**2.2 hours**

Number of women serving in managerial/ leadership /ownership role for the group of VPA's[47]

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0	300	300
Year 2	0	900	900
Year 3	0	1,200	1,200
Year 4	0	1,200	1,200
Year 5	0	1,200	1,200
Year 6	0	1,200	1,200
Year 7	0	1,200	1,200
Year 8	0	1,200	1,200
Year 9	0	1,200	1,200
Year 10	0	1,200	1,200
Total	**0**	**10,800**	**10,800**
Total number of crediting years	10		
Annual average over the crediting period	0	1,080	1,080

Ex-ante estimation of SDG 7 Outcome

Number of households equipped with F3PA cookstoves for the group of VPA's[48]

Year	Baseline estimate	Project estimate	Net benefit

[47] The following figures are evaluated based on an implementation plan of 1 VPA in Year 1, 2 in Year 2 and 1 VPA in Year 3. It is estimated that VPAs are covered at 60% (or 1,440 households) in their first year of implementation.
[48] The following figures are evaluated based on an implementation plan of 1 VPA in Year 1, 2 in Year 2 and 1 VPA in Year 3. It is estimated that VPAs are covered at 60% (or 1,440 households) in their first year of implementation.

Year 1	0	1,680	1,680
Year 2	0	6,160	6,160
Year 3	0	10,080	10,080
Year 4	0	11,200	11,200
Year 5	0	11,200	11,200
Year 6	0	11,200	11,200
Year 7	0	11,200	11,200
Year 8	0	11,200	11,200
Year 9	0	11,200	11,200
Year 10	0	11,200	11,200
Total	**0**	**96,320**	**96,320**
Total number of crediting years	10		
Annual average over the crediting period	**0**	**9,632**	**9,632**

Ex-ante estimation of SDG 15 Outcome

Quantity of non-renewable wood saved due to the use of F3PA cookstoves for group of VPA'

Year	Baseline estimate t of wood	Project estimate t of wood	Net benefit t of wood
Year 1	5,336	2,627	2,709
Year 2	19,565	9,669	9,896
Year 3	32,016	15,989	16,026
Year 4	35,573	17,972	17,601
Year 5	35,573	18,129	17,444
Year 6	35,573	18,287	17,286
Year 7	35,573	18,447	17,126
Year 8	35,573	18,609	16,965
Year 9	35,573	18,772	16,801

Year 10	35,573	18,937	16,637
Total	**305,930**	**157,438**	**148,492**
Total number of crediting years	10		
Annual average over the crediting period	**30,593**	**15,744**	**14,849**

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 1

Data / Parameter	Average household annual savings i.e., decrease in expenditure on wood fuel purchase Various expenses which women do after saving money required for the purchase of wood: (i) School fees$_p$; (ii) Purchase of medical drugs$_p$; (iii) Purchase of food$_p$; (iv) Investment for field crops$_p$; (v) Purchase of equipments$_p$; (vi) Income generating activities$_p$; (vii) Savings$_p$
Unit	€ or FCFA
Description	Total estimated amount in FCFA or € saved by the stove users on wood fuel purchase during the monitoring period (i) School fees$_p$: Proportion of stove users using their saved money to school fees; (ii) Purchase of medical drugs$_p$: Proportion of stove users using their saved money to purchase of medical drugs; (iii) Purchase of food$_p$: Proportion of stove users using their saved money to (iv) Investment for field crops$_p$: Proportion of stove users using their saved money to investments for field crops; (v) Purchase of equipments$_p$: Proportion of stove users using their saved money to purchase of equipments;

	(vi) Income generating activities$_p$: Proportion of stove users using their saved money to income generating activities; (vii) Savings$_p$: Proportion of stove users using their saved money to savings.
Source of data	Monitoring surveys
Value(s) applied	7 €
Measurement methods and procedures	The measurement of the parameter is based on quantitative information collected during Monitoring surveys. The end users are asked whether they purchase wood fuel and if so, how much they spend on yearly basis.
Monitoring frequency	Annual
QA/QC procedures	The data will be analysed in the monitoring report and raw data of the monitoring surveys will be made available for review.
Purpose of data	Calculation for the parameter "Average household annual savings i.e., decrease in expenditure on wood fuel purchase"
Additional comment	N/A

Data / Parameter	Total estimated amount saved by stove users on wood fuel purchase for the group of VPA's
Unit	€ or FCFA
Description	Total estimated amount in FCFA or € saved by the stove users on wood fuel purchase during the monitoring period for the group of VPA's
Source of data	Monitoring surveys
Value(s) applied	€ 13,558
Measurement methods and procedures	The measurement of the parameter is based on quantitative information collected during Monitoring surveys. The end users are asked whether they purchase wood fuel and if so, how much they spend on yearly basis. The result is extrapolated is on all stove users within the VPA.

Monitoring frequency	Annual
QA/QC procedures	The data will be analysed in the monitoring report and raw data of the monitoring surveys will be made available for review.
Purpose of data	Calculation for the parameter "Total estimated amount saved by stove users on wood fuel purchase"
Additional comment	N/A

SDG 3

Data / Parameter	Smoke level reduction Incidence of coughing reduction Incidence of respiratory illness reduction Incidence of itchy eyes reduction
Unit	Fraction
Description	Proportion of households perceiving less often smoke levels, incidence of coughing, incidence of respiratory illness, incidence of itchy eyes since the implementation of F3PA efficient cookstoves
Source of data	Monitoring surveys
Value(s) applied	90% 90% 90% 90%
Measurement methods and procedures	The measurement of the parameter is based on qualitative information collected during monitoring surveys. The end users are asked whether, since they have the F3PA efficient cookstoves, smoke level occurs for each more often, less often among the family members or the situation has not changed. The same is asked for coughing, respiratory illnesses and itchy eyes.
Monitoring frequency	Annual

QA/QC procedures	The data will be analyzed in the monitoring report and raw data of the monitoring surveys will be made available for review.
Purpose of data	Calculation of the parameter "Proportion of households perceiving less often smoke levels, incidence of coughing, incidence of respiratory illness, incidence of itchy eyes"
Additional comment	Based on previous monitoring reports of other projects of Association tiipaalga: https://impact.sustain-cert.com/document_files/25599 page 14

SDG 4

Data / Parameter	Number of workshops carried out for women per group of VPA's per year
Unit	Number
Description	Number of workshops carried out for women in order to increase their empowerment
Source of data	Reports regarding the workshops carried out for women
Value(s) applied	72
Measurement methods and procedures	The list of workshops carried out for women during the corresponding monitoring period
Monitoring frequency	Annual
QA/QC procedures	The data will be analyzed in the reports regarding the workshops carried out for women, which will be made available for review.
Purpose of data	Calculation of the parameter "Number of workshops carried out for women"
Additional comment	N/A

Data / Parameter	Number of employees provided skill development training
Unit	Number

Description	Number of employees involved in the program who received training to increase their performance for their role within the project.
Source of data	Reports regarding the training initiatives
Value(s) applied	5.3 8 employees in Year 1 5 employees subsequent years
Measurement methods and procedures	The list of training initiatives during the corresponding monitoring period
Monitoring frequency	Annual
QA/QC procedures	The data will be analyzed in the reports regarding the training initiatives, which will be made available for review.
Purpose of data	Calculation of the parameter: "Number of employee provided skill development training"
Additional comment	N/A

SDG 5

Data / Parameter	Average time saving associated with cooking time and fuel collection per woman per week
Unit	Hours/week
Description	Average time saving (in hours) on wood fuel collection per woman per week during the monitoring period i. Domestic tasks$_p$ = Proportion of stove users using their saved time to do domestic tasks ii. Income generating activities$_p$ = Proportion of stove users using their saved time to do income generating activities iii. Field labour$_p$ = Proportion of stove users using their saved time to do field labour iv. Gardening$_p$ = Proportion of stove users using their saved time to do gardening v. Participation to a literacy program$_p$ = Proportion of stove users using their saved time to participate to a literacy program

	vi. Community $work_p$ = Proportion of stove users using their saved time to do community work vii. Religious $activities_p$ = Proportion of stove users using their saved time to do religious activities
Source of data	Monitoring surveys
Value(s) applied	2.2 hours
Measurement methods and procedures	The measurement of the parameter is based on quantitative information collected during monitoring surveys. The end users are asked, how much time they spent collecting wood fuel for domestic cooking since they have the F3PA efficient cookstoves.
Monitoring frequency	Annual
QA/QC procedures	The data will be analysed in the monitoring report and raw data of the monitoring surveys will be made available for review.
Purpose of data	Calculation of parameter: "Average time saving associated with cooking time and fuel collection per woman per week"
Additional comment	N/A

Data / Parameter	Number of women serving in managerial/leadership /ownership role for the group of VPA's
Unit	Number
Description	Number of leader women which will be formed by tiipaalga staff to teach the construction methods for the F3PA cookstoves. The trained women will then form other women in the rural villages so that they will implement the F3PA technology within their households.
Source of data	Reports regarding the workshops carried out for women
Value(s) applied	1,080 leader women per year
Measurement methods and procedures	The list of workshops carried out for women during the corresponding monitoring period
Monitoring frequency	Annual

QA/QC procedures	The data will be analyzed in the reports regarding the workshops carried out for women, which will be made available for review.
Purpose of data	Calculation of the parameter "Number of women serving in managerial/ leadership /ownership role for the group of VPA's"
Additional comment	N/A

SDG 7

Data / Parameter	Number of households equipped with F3PA cookstoves for the group of VPA's
Unit	Number
Description	Number of households included in the project database that are equipped with F3PA cookstoves
Source of data	Project database
Value(s) applied	9,632
Measurement methods and procedures	The project database provides a list of end-users with number of F3PA efficient cookstoves per end-user.
Monitoring frequency	Continuous
QA/QC procedures	The data will be analysed in the monitoring report and project database will be made available for review.
Purpose of data	Calculation of parameter: "Number of households equipped with F3PA cookstoves for the group of VPA's"
Additional comment	N/A

SDG 13

Data / Parameter	$U_{p,i}$
Unit	Percentage
Description	Usage rate in project scenario p during year i
Source of data	Annual usage/monitoring survey

Value(s) applied	95%
Measurement methods and procedures	The measurement of the usage rate is based on qualitative information collected in the usage/monitoring survey. The measurement methods and procedures in alignment with The Gold Standard Simplified Methodology for Efficient Cookstoves Section C. are described further in Section B.7.3. A question concerning the current use of the technology is asked to each end-user of the sample and is validated by the observation of the surveyor in order to determine the usage rate of each technology age category. The proposed ex-ante values are aligned with estimates used for other projects under the PoA GS1340.
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of emission reductions
Additional comment	N/A

Data / Parameter	$N_{p,i}$
Unit	Number of households included in the project (Units), based on days of usage of age group i during the corresponding monitoring period related to one year.
Description	Household in the project database for project scenario p through year i for which all baseline cookstove set(s) (comprising of several traditional three stone cookstoves for domestic use) have been replaced by project cookstove set(s)
Source of data	Project database
Value(s) applied	2,800
Measurement methods and procedures	For the determination of the number of usage days at household level for age group I during the corresponding monitoring period, the latest start day of use of all constructed F3PA efficient cookstoves within the household will be taken in order to have conservative approach.

	Number of households included in the project (Units) are calculated based on days of usage of age group I during the corresponding monitoring period related to one year.
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of emission reductions
Additional comment	N/A

Data / Parameter	DF_{η}
Unit	Fraction
Description	Discount factor to account for efficiency loss of project stoves
Source of data	Gold Standard Simplified Methodology for efficient Cookstoves
Value(s) applied	Default value: 0.99 i.e 1% efficiency loss/year
Measurement methods and procedures	N/A
Monitoring frequency	N/A
QA/QC procedures	N/A
Purpose of data	Calculation of emission reductions
Additional comment	N/A

Data / Parameter	$DF_{b,stove,i}$
Unit	Percentage
Description	Discount factor to account for the baseline stove use in project scenario p during year i
Source of data	Monitoring surveys
Value(s) applied	2%
Measurement methods and procedures	The measurement of the discount factor to account for the baseline stove use is based on qualitative information

	collected in the usage/monitoring survey. A question concerning the current use of the baseline technology is asked to each end user of the sample and is validated by the observation of the surveyor in order to determine the discount factor to account for the baseline stove use in project scenario p of each technology age category.
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of emission reductions
Additional comment	N/A

SDG 15

Data / Parameter	Quantity of non-renewable wood saved due to the use of F3PA cookstoves for group of VPA's
Unit	T of wood
Description	Quantity of tons of wood saved due to the use of the F3PA cookstove during the monitoring period
Source of data	Calculated
Value(s) applied	14,849
Measurement methods and procedures	The calculation is done as described in section B.6.1
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of saved wood
Additional comment	N/A

B.7.2. Sampling plan

As described in the PoA-DD cross sampling across a group of VPA's is allowed if the VPA's are homogeneous relative to the parameters of interest. Cross sampling of households will be applied across the following VPAs of PoA 1340: VPA-25 (GS11070), VPA-26 (GS11071), VPA-27 (GS11072), VPA-28 (GS11073). The parameters which need to be monitored through surveys for the 4 VPA's are (i) Up,y Usage rate in project scenario p during year y; (ii) DFη Discount factor to account for efficiency loss of project stoves; and (iii) DFb,stove,y Discount factor to account for the baseline stove use in project scenario p during the year y. The 4 VPA's included are assumed to be homogeneous with respect to the three parameters for the following reasons:

i. the 7 municipalities included in the 4 VPAs are very close to each other (furthest distance of 110 km);
ii. the intention is to implement the 4 VPAs within the timeframe of 3 years (1 VPA in Year 1, 2 VPA's in Year 2 and 1 VPA in Year 3);
iii. all 4 VPAs will apply the same distribution/implementation mechanism. The distribution/implementation mechanism involves training local women in the rural zones to construct these stoves themselves using local material. Each efficient cookstove will be built according to a strict construction protocol. The efficient cookstove construction instructions are published in the training material provided and all levels of the distribution process will be trained to have full knowledge of this construction criterion. This innovative distribution system includes a tight collaboration with women associations. The construction protocol required to train the local rural inhabitants will give them the skills they require to build their own personally fabricated standardized efficient F3PA cookstove under the supervision of tiipaalga and how to use and maintain it;
iv. the technology used in all 4 VPA's will be the same, the standardized efficient F3PA cookstove or the mud made 3 stones efficient cookstove constructed according to a strict construction protocol. In all 4 VPA's the household will have the choice between different sizes according to their needs. As all levels of the distribution mechanism are trained to construct according to the strict construction protocol, the efficiency will be identical;

Since the three parameters of interest are assumed to be the same in each VPA at the time of sampling survey during the monitoring period, a single survey with cross sampling of households can be undertaken using a single random sampling plan. The populations of all 4 VPAs are combined together and then the sample size is calculated using the sampling guidelines described below.

The number of households of which each wife of the household (when polygamous) has replaced all traditional three stones cookstoves for domestic use with project cookstoves, is recorded in the project database. Only the households recorded in the database will be part of the project activity.

To successfully conduct a usage survey, the minimum household sample size of each age-group should be in line with the following guidelines (according to the Gold Standard Simplified Methodology for Efficient Cookstoves):

- Project target population < 300: Minimum sample size 30;
- Project target population 300 to 1000: Minimum sample size 10 % of group size;
- Project target population > 1000: Minimum sample size 100.

The method of selecting households for the sample list for the monitoring survey will be random. All households will according to the installation date of the project cookstoves be assigned to a specific age-group. Per age-group households will be selected using a random sequence generator (*Rand()*) function in excel or other random sequence generator[49]. All random selections will be stored for the crediting period and an additional two years, which allow traceability of the selection.
For all parameters that are monitored via sampling it is understood that only the age of the project cookstove has an influence. Therefore, no geographic representativeness is deemed necessary for the selection of users participating in the sample groups.
The periodical checks will be performed by user interviews. Only persons older than 18 years will be interviewed.

B.7.3. Other elements of monitoring plan

Initial Data Collection

The initial data collection serves as the foundation of the project monitoring plan. Each stove is built by the user in collaboration with a local leader woman selected by the women in the villages, and who is trained by APEDR facilitators with the supervision of CO2logic and Solidagro.
The mud made 3 stones efficient woodstove or efficient F3PA cookstoves have a strict construction protocol[50] with diagrammatic presentation material[51] to support the APEDR trainers, skilled by Association tiipaalga, and the leader women on the ground and ensure that project stoves are systematically built in a similar manner. They will also train stove users/owners on how to use and maintain the efficient cookstoves and educate on hygienic usage and on the threat of climate change and health hazards related to the old cooking system.
Some of the households in the project area of the VPA are polygamous. Each wife of the household included in the carbon project must have a cookstove set of at least two efficient F3PA cookstoves of different sizes. The project cookstoves are single pot stoves. As every cooking pot size has its specific size of cookstove, different sizes of project cookstoves will be implemented according to the cooking habits of the stove users. The sizes of the cooking pots and so the cookstoves used in this VPA are 2, 3, 4, 5, 6, 7, 8, 10, 12 and 15 due to their frequency of utilization. The women using different cookstove sets in a polygamous household are credited as one single household.
The individual identification of this micro scale - VPA is ensured with the identification of each household and each wife within the household using the project cookstoves by a unique serial number referring to this micro scale – VPA.

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[49] For example: https://www.randomdraws.com/random-sequence-generator/
[50] See document « Fiche technique de construction de F3PA_tiipaalga »
[51] See documents « tiipaalga -Boite à images » and document « Guide d'animation_tiipaalga_vf »

The following information will be documented for each household of which each wife of the household (when polygamous) has replaced all traditional three stones cookstoves for domestic use with project cookstoves:

i. Unique VPA ID- number of each household and each wife within the household;
ii. Type and size of appliance (ex. F3PA – size 2);
iii. GPS Coordinates of the household;
iv. Name/Address/national ID Number/Mobile Number/Picture of wife with her project cookstoves;
v. Stove Construction Date;

The data collected by APEDR and their team on the ground will be uploaded to a central database online. The collection of each component is briefly described below.
As there is only one project scenario the project database does not need to be differentiated into different sections.

Unique VPA ID- number of each wife of the household
Each wife of the household will receive a unique serial number. The syntax of the unique serial number is defined as GS1340-VPA-25-xxxx/y where (i) GS1340 is the Gold Standard number of the PoA "Efficient cookstoves in Burkina Faso" to which the VPA belongs, (ii) VPA-25 is the number of the VPA of the PoA, (iii) xxxx is the number of the household from 1 to 9999 and (iv) y is the number of the wife in the household from 1 to 9.

Type and size of appliance
In this VPA one type of appliance will be deployed, which is the mud made 3 stones efficient woodstove. However different sizes of cookstoves can be constructed at the level of the households depending on their cooking habits and size of household.

GPS
After the installation of all project cookstoves within a household, a APEDR team member will register the GPS coordinates of the household.

Name/Address/national ID Number/Mobile Number/Picture of wife with her project cookstoves
Each wife of the household participating in the project will use at least two project cookstoves. The unique VPA ID-number will identify the household and the wife included in the VPA-25. To ensure further traceability personal information of the stove recipient will be recorded. Since rural households in project locations do not have an official address, location data may be limited to the name of the village. In addition to the GPS location and the village name, a picture of each stove user with her project cookstoves will be taken. If available and if the concerned person wants to share, the national ID number and mobile phone number of the husband and each wife of the household will be collected. APEDR will strive to obtain as much unique information regarding the household and each wife of the household as possible.

Stove Construction Date
The construction date of each stove will be recorded during the initial data collection. The construction date will be uploaded to the electronic database containing the previously described stove information.
The contract concerning the transfer of carbon rights between the user and project developers will be signed by the stove user after construction and positioning in the kitchen of at least two efficient cookstoves per wife of each household. The hardcopies of the contracts will be archived electronically at the office of Solidagro.
A household for which all of this data has been collected for each wife of the household and uploaded to the electronic database is deemed to have completed the initial data collection. The virtual copy of the stove and household information will be stored for the crediting period and an additional two years.

Monitoring Plan

In accordance with the Gold Standard Simplified Methodology for Efficient Cookstoves, the following data will be monitored over the crediting period of the project activity:

i. Usage rate in project scenario p during year y $U_{p,y}$ (%)
ii. Number of project cookstoves credited (units), $N_{p,y}$
iii. Discount factor to account for efficiency loss of project cookstoves DF_{η}
iv. Discount factor to account for the baseline stove use in project scenario p during the year y, DF_{p}, stove y

Monitoring consists of checking of a representative sample for each age-group of project cookstoves installed for each wife of a household included in the VPA, once every year (annually) to ensure that project cookstoves are still operating by carrying out the usage survey.
A usage survey will be conducted to estimate the drop off rates at household level as the project cookstove of one of the wives of the household may not be adopted or may be disposed of and potentially replaced again by a baseline stove. Prior to the verification, a usage survey for each cookstove age-group is required. For example, if only cookstoves in the first year of use (age 0-1) are being credited, a usage parameter must be established for age-group 0-1, through a usage survey for cookstove age 0-1. If cookstoves of age 0-1 and age 1-2 are being credited (as part of first request of issuance), usage parameters must be established for age-group 0-1 and 1-2, respectively through a usage survey. If cookstoves of age-group 0-1 and age-group 1-2 are being credited (as part of second request for issuance), usage parameter must be established for age-group 1-2 only through a usage survey as the usage rate for cookstoves of age group 0-1 can be applied from the previous issuance.

Usage rate in project scenario p during year y $U_{p,y}$ (%)
From the monitoring survey, a usage rate parameter (%) is derived from each age group of project cookstove installed for each wife of a household included in the VPA.
During the survey, the project cookstove will be checked if it is in useable condition. If the project cookstove is not in useable condition, the household to which the project

cookstove belongs will be excluded from the project database for the whole crediting year and subsequent years. The household will be included again after repairing or replacing it with new cookstove with similar efficiency. Guidance provided in Annex B of the methodology 'The Gold Standard Simplified Methodology for Efficient Cookstoves' will be followed to evaluate the condition of the cookstoves.
Cookstove set(s) within a household can only be considered 'in use' if all the cookstoves in the set(s) (in polygamous households all cookstoves of all cookstove sets of all women in the household) are being used. Similarly, cookstove set(s) can only be considered in 'good condition' as long as all cookstoves within the cookstove set(s) (in polygamous households all cookstoves of all cookstove sets of all women in the household) are in a 'useable condition'.

Number of households credited (units), $N_{p,y}$
This is the number of households from the project database based on the number of usage days per age group.

Discount factor to account for efficiency loss of project cookstoves DF_{η}
The default value of 1% efficiency loss per year can be used if the stove is found in good condition during annual survey.
For each year, the stoves of the age-group x-y should be physically verified. In the case of progressive installations, stove of age-group 0-1 shall also be physically verified each year through a random sampling approach.
During annual surveys, if it is found that the project cookstoves of some households are not in working conditions, the proportionate population of households should be excluded from the project database, until these cookstoves are repaired or replaced with new cookstoves. A site visit by an Objective Observer with relevant technical background would be required at the time of first internal verification and then subsequently after every year from the previous issuance.

Discount factor to account for the baseline stove use in project scenario p during the year y, DF_p, stove y
This parameter will be determined based on number of meals cooked using the baseline stove. The required information shall be captured through sample surveys carried out following a random sampling approach for each age-group of the project stove.
In case of polygamous households the discount factor shall be determined for each cookstove set and the highest value of all cookstove sets within the household shall be used as representative discount factor for the household.

Database records

Electronic database(s) will be operated and maintained by APEDR and Solidagro with the technical support of CO2logic to ensure completeness and accuracy of monitoring information.

Project database[52]:

- Unique VPA ID-number of each wife of the household;
- Type and size of appliance (ex. F3PA – size 2);
- GPS Coordinates of the household;
- Name/Address/national ID Number/Mobile Number/Picture of wife with her project cookstoves;
- Stove Construction Date;
- Status of stoves owned by the wife: used, unused, destroyed or replaced.

The information in this database will be updated continuously.
Sample database:

- Unique ID-number of the household
- Household profile
- Fuel consumption pattern post project implementation per wife of the household with unique VPA-ID-number:
    - Cooking device
    - Place for cooking
    - Type of fuel and fuel consumption

The information in this database will be updated for every monitoring period. Data will be collected with Smart phones and transferred to the electronic database.

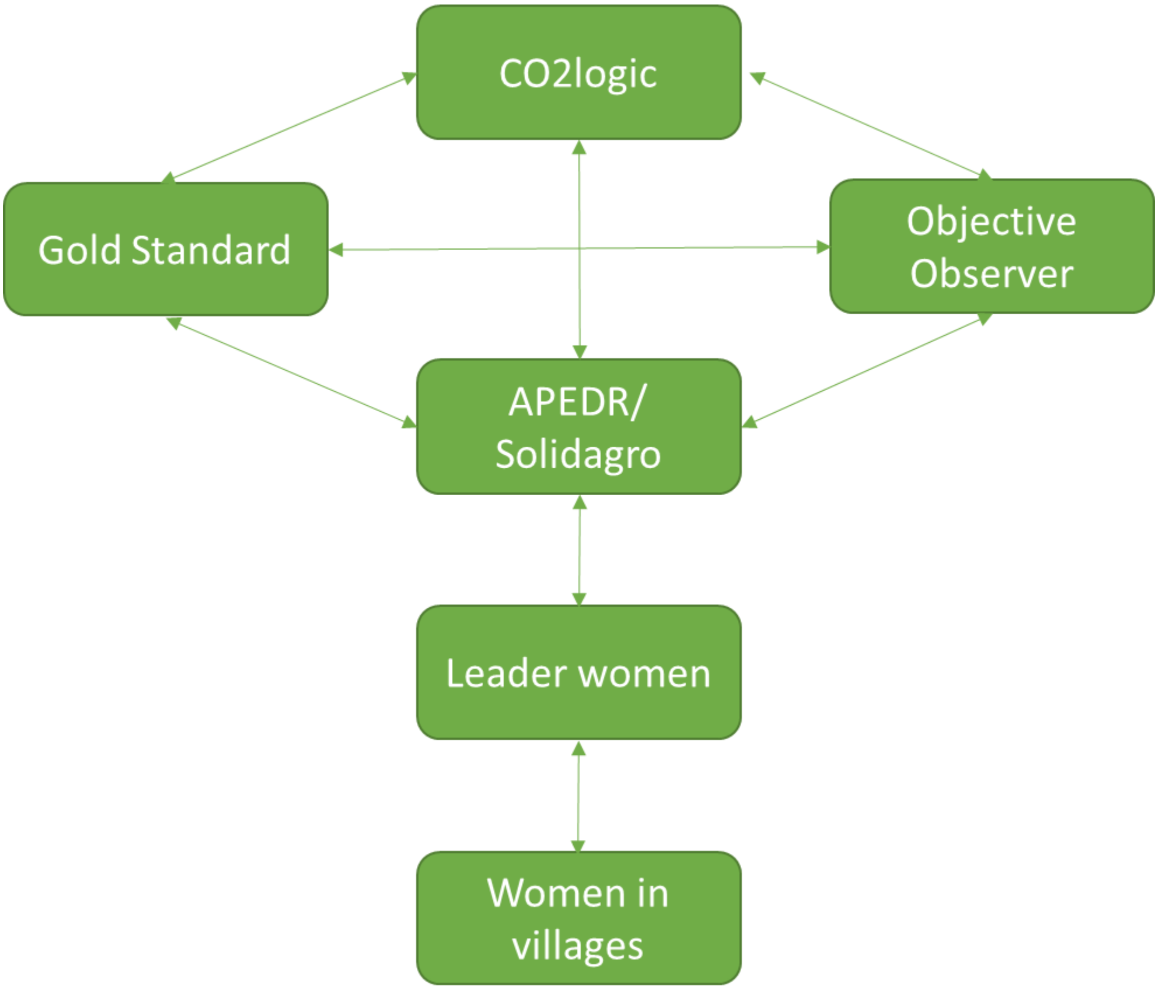
Monitoring Report

One single monitoring report will be written for the group of 4 VPA's at the end of every verification period and submitted to the SustainCERT. The report will indicate how the monitoring data has been collected and show detailed, conservative calculations of the emissions reductions for the verification period and project in question.
The initial and monitoring data for each verification period stored in the electronic database will serve as the backbone of the monitoring report. The report will contain extensive tables comparing the initial data collected during project implementation with the monitoring data. This allows for quick confirmation that the project cookstoves of each wife of the household in question are still operational.
For replaced project cookstoves, the initial data collection process will be repeated and uploaded to the electronic database.

[52] The record keeping system should collect as many information as necessary to facilitate the verification of the VERs. At the current point of time the list of information seems ideal but may be extended or condensed. The collection of all the items is therefore not mandatory and additional information may be collected as well.

Diagram of Responsibilities

As there are several entities involved in initial data collection and project monitoring it is important to clearly designate the relationships between and responsibilities of entities. CO2logic will act as the managing entity of the project and be responsible for communication with the Gold Standard Foundation and the Objective Observer. A diagram of responsibilities is shown here below.



APEDR employees, supervised by Solidagro and CO2logic, train leader woman who are selected by the women in the villages, in the construction, the use and maintenance of mud made 3 stones efficient woodstoves. These leader women conduct the same training sessions with the women in their villages and help them to build the cookstoves. APEDR employees in collaboration with the leader women will perform quality checks and collect the initial stove data described earlier.
The collected data will be transferred electronically from the project site to the office of Solidagro. The initial project data will be processed and uploaded to the central electronic database accessible by project participants and SustainCERT. CO2logic will perform quality checks.
For project monitoring the APEDR trainer in collaboration with the leader women will revisit the project site to monitor a representative sample of the project activity.

Monitoring data will be collected and processed in the way the initial data was collected and processed.
APEDR with technical assistance of CO2logic and Solidagro will provide training to parties involved in the monitoring plan to assure accuracy and completeness of data recorded. The trainings will be conducted at the time when it is most appropriate during the project implementation phase.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of project

10/03/2021[53]

C.1.2. Expected operational lifetime of project

10 years.

C.2. Crediting period of project

C.2.1. Start date of crediting period

25/03/2021[54]

C.2.2. Total length of crediting period

10 years[55].

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in Appendix 1, ongoing monitoring is summarised below. No mitigation measures have been identified in the safeguarding principles assessment.

Principles	Mitigation Measures added to the Monitoring Plan
Principle x.y	

[53] See signed contract: 'Carbon waiver - VPA 25 - HH 35_with comments'. The construction date in the contract is 10/03/2021, whereas the installation date (registered in project database) is 25/03/2021, which is 2 weeks after the construction date. The installation date is to be considered as the first date of usage of the F3PA efficient cookstove and thus start date of crediting period.

[54] See signed contract: 'Carbon waiver - VPA 25 - HH 35_with comments'. The construction date in the contract is 10/03/2021, whereas the installation date (registered in project database) is 25/03/2021, which is 2 weeks after the construction date. The installation date is to be considered as the first date of usage of the F3PA efficient cookstove and thus start date of crediting period.

[55] Rule update: For PoAs any VPA submitted within the first crediting period (i.e. 7 years) of the PoA shall be allowed to use the crediting period of 10 years. The PoA "Efficient cookstoves in Burkina Faso" is still in its first crediting period.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The Project takes into account gender roles and the abilities of women and men to participate in the decision/designs of the project activities. For example, the stakeholder consultation made in the project design phase will include both women and men participating in the consultation meeting. Moreover, for example, the future public awareness sessions and trainings for the construction of F3PA efficient cookstoves will be planned and organized in the way to avoid any discrimination of women or other marginalized groups. In fact, the women's participation will be essential for guaranteeing the success in the dissemination of the F3PA efficient cookstoves. In the overwhelming majority of households in Burkina Faso, fuel collection activities are handled by women. In fact, the reduction of fuel wood consumption will significantly reduce women's workload related to collection of fuel. It can be further expected that sexual harassment and violence happening during fuel collection may be reduced. Hence, largely women will benefit from the project activity.	

Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The Constitution of Burkina Faso[56] recognizes that the promotion of gender is a factor in achieving equality between men and women in Burkina Faso. In July 2009, the government adopted a National Gender Policy (PNG). The vision entails the construction of "a society free of all forms of inequality and inequities of gender, and which ensures, for all its citizens, the necessary safeguards for their social, cultural, political and economic development[57]. It will be ensured that the project is committed to equal gender rights following the National Gender Policy of Burkina Faso.	
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No expert is required to assess the Gender Safeguarding Principles & Requirements as gender has been adequately assessed in the Safeguarding Principles Assessment.	
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No expert is required to assist with gender issues at the stakeholder consultation as gender has been adequately assessed in the Safeguarding Principles Assessment.	

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

56 http://www.accpuf.org/images/pdf/cm/burkinafaso/031-tf-txt_const.pdf : page 2

57 https://www.ilo.org/dyn/natlex/docs/SERIAL/90589/104519/F-957349365/BFA-90589.pdf : page 4

On the 25th of February 2021, the local stakeholder consultation for the project "Solidagro F3PA cookstoves in Passoré" took place in the conference room of the Youth House in Yako. The project was presented to the different stakeholders. During the discussion phase, they were invited to address their questions over the different aspects of the project. Different comments from the consultations were addressed and answered at that time. Individuals who could not attend the meeting were invited to raise their queries through email, telephone or fill in the grievance books at APEDR office in Yako or at Solidagro's office in Ouagadougou. No other comments were received.
Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

All the questions raised by the stakeholders during the consultation have been discussed with the participants and project developers and answered. In addition, no grievance nor comment were formulated during the Stakeholders Feedback Round (SFR) conducted between the 24/06/2021 and 31/08/2021[58].
As no major negative comment has been suggested, no stakeholder mitigation measures have been reported.
The stakeholder's comments did not lead to major changes in the project design; this is probably because the project has been designed in partnership with the local communities from the beginning.

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	Grievance book is available at APEDR office in Yako. Another book is made available at Solidagro's office in Ouagadougou.
GS Contact (mandatory)	help@goldstandard.org
Telephone access	Alain Touta Traoré/ Representative of Solidagro in Burkina Faso +226 70 70 61 29 Sayouba Bonkoungou/ President of APEDR +226 76 50 86 42

58 See Stakeholders Consultation Report Section E)

Email Access	info@solidagro.be

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into SECTION D above. Please refer to the instructions in the Guide to Completing this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights	No The project is implemented on the ground by Solidagro, an agro-ecology and climate NGO, the Association for the Protection of the Environment and Rural Development (APEDR) and CO2logic, a Belgian consultancy company for the technical assistance with the carbon market aspects. The project developers take care that the project respect internationally	N/A	N/A

	proclaimed human rights and is no complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights. In addition, the Republic of Burkina Faso has acceded to the Human Rights Convention under the United Nations on 4th of January 1999[59].		
2. The Project shall not discriminate with regards to participation and inclusion	No All households located within the project boundary that wish to have the F3PA efficient cookstove, are able to do so and there is not any form of discrimination or exclusion to participate in the project. The Project Developers follow the	N/A	N/A

[59] https://treaties.un.org/pages/ViewDetails.aspx?src=TREATY&mtdsg_no=IV-4&chapter=4&clang=_en

	Non-Discrimination policy of Fairtrade International.		
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women	No The Project will not directly or indirectly reinforce gender-based discrimination and will not lead to /contribute to adverse impacts on gender equality or the situation of women, like sexual harassment and/or any forms of violence against women - address the multiple risks of gender-based violence, including sexual exploitation or human trafficking. The Project will not directly or indirectly reinforce gender-based discrimination and will not and will not lead to/contribute to adverse impacts on gender equality or the situation of women and not to lead to any risk of contributing issues like	N/A	N/A

	slavery, imprisonment, drudgery or coercion of women and girls. The project will not restrict women's rights or access to natural resources. The project will not have any impact on women's ownership rights to inherit and own land, homes and other assets.		
2. Projects shall apply the principles of nondiscrimination, equal treatment, and equal pay for equal work	No The Project applies the principles of non discrimination and equal treatment. For any other eventual paid or volunteer work the principle of the equal pay for equal work will be applied and organized in way to provide the conditions for equitable participation of men and women.	N/A	N/A

	Project activities are planned in way that the participation of both women and men is guaranteed. There are no limitations on participation or benefiting from the Project depending on the pregnancy, maternity/paternity leave or marital status.		
3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks	No The Project will not include any specific gender related risks. Burkina Faso adopted a National Gender Policy in July 2009[60]. The vision that underpins this policy is the construction of a "society free of all forms of gender inequalities and inequities,	N/A	N/A

60 http://extwprlegs1.fao.org/docs/pdf/bkf146346.pdf

	and which provides all its citizens with the security essential for their social, cultural, political and economic development. It will be ensured that the project is committed to equal gender rights following article 1 of the constitution[61].		
4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	N/A	N/A	N/A
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	No The project activity does not expose the community to increased health risks and is not adversely affecting the health of workers and the community.	N/A	N/A

61 https://www.constituteproject.org/constitution/Burkina_Faso_2012.pdf

	More in general the project participants follow the national regulations of Burkina Faso on health, safety working conditions.		
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No The efficient F3PA project cookstoves will not critically alter cooking practices. So it has no impact on cultural heritage.	N/A	N/A
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No The project does not lead to relocation of people.	N/A	N/A
>>			
Principle 4.3 Land Tenure and Other Rights			
Does the Project require any change to land tenure arrangements and/or other rights?	No The project does not require any change in land tenure arrangements and/or other rights.	N/A	N/A
>>			

Principle 4.4 Indigenous Peoples			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples? >>	No There are no indigenous people (referring as distinct collectives) present in the project area.	N/A	N/A
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No The Project does not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects. The project is implemented on the ground by APEDR. The ethical codes of APEDR is against corruption. Moreover, Burkina Faso has ratified the UN Convention against Corruption.	N/A	N/A
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour	No	N/A	N/A

occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions.	The project does not involve and is not complicit in any form of forced or compulsory labour. All labour is voluntary and APEDR does not engage in any form of forced, compulsory or child labour. There won't be any form of labour discrimination. Labour conditions for workers will be safe. Burkina Faso is member state of the International Labour Organization (ILO)[62]. The country bases its legal regulations on the ratified ILO conventions[63].		

62 http://www.ilo.org/public/english/standards/relm/country.htm

63 http://www.ilo.org/global/standards/introduction-to-international-labour-standards/conventions-and-recommendations/lang--en/index.htm

2. Workers shall be able to establish and join labour organisations	No All the stakeholders involved in the project are totally free to associate with whomever they want.	N/A	N/A
3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave.	No The working agreements with the individual workers are documented and implemented and the minimum requirements stated in section 3.6.1. of GS4GG Safeguarding Principles & Requirements (version 1.1) are respected whenever applicable.	N/A	N/A

4. No child labour is allowed (Exceptions for children working on their families' property requires an [Expert Stakeholder](#) opinion)	No All the possible staff hired by APEDR has a minimum age of 18. Burkina Faso has ratified ILO Conventions 138 (Minimum Age Convention) and 182 (Worst Forms of Child Labour Convention).[64]	N/A	N/A
5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures	No All the works will be made by using appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures.	N/A	N/A
Principle 6.2 Negative Economic Consequences			

64 https://www.ilo.org/dyn/normlex/en/f?p=1000:11200:0::NO:11200:P11200_COUNTRY_ID:102981

1. Does the project cause negative economic consequences during and after project implementation?	No Project activity related costs, like e.g. for construction and maintenance of F3PA efficient cookstoves, monitoring etc are covered with climate finance and carbon finance. The use of the F3PA efficient cookstoves is accessible to everybody and therefore the project benefits can be considered socially-inclusive. Hence, the project causes no negative economic consequences during and after project implementation, e.g. for vulnerable and marginalised social groups in targeted communities.	N/A	N/A
>>			
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No On the contrary, the project will reduce greenhouse gas	N/A	N/A
>>			

	emissions over the baseline scenario.		
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users? >>	No On the contrary, the objective of the project is to reduce the consumption of wood.	N/A	N/A
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity? >>	No The efficient F3PA project cookstoves will lead to decreased deforestation, decreased soil erosion and thus decreased sedimentation in lakes and rivers, and increased infiltration of water in the soil.	N/A	N/A
Principle 8.2 Erosion and/or Water Body Instability			
Could the Project directly or indirectly cause additional erosion and/or water	No	N/A	N/A

body instability or disrupt the natural pattern of erosion?	The efficient F3PA project cookstoves will lead to decreased deforestation, decreased soil erosion and thus decreased sedimentation in lakes and rivers, and increased infiltration of water in the soil.		
>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No The Project does not use land and soil for the production of crops or other products.	N/A	N/A
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No The Project will not be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions.	N/A	N/A
>>			

Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)? >>	No The Project does not lead to the use of genetically modified organisms.	N/A	N/A
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment? >>	No The Project does not result in the release of pollutants to the environment.	N/A	N/A
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials? >>	No The Project does not involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials.	N/A	N/A
Principle 9.6 Pesticides & Fertilisers			

Will the Project involve the application of pesticides and/or fertilisers? >>	No The Project does not involve the application of pesticides and/or fertilizers.	N/A	N/A
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests? >>	No The Project does not involve the harvesting of forests.	N/A	N/A
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives? >>	No The Project does not modify the quantity or nutritional quality of food.	N/A	N/A
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry? >>	No The Project does not involve animal husbandry.	N/A	N/A
Principle 9.10 High Conservation Value Areas and Critical Habitats			

Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified? >>	No The introduction and usage of improved cook stoves will not physically affect or alter largely intact or HCV ecosystems, critical landscapes or key biodiversity areas or sites in the region.	N/A	N/A
Principle 9.11 Endangered Species			
Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? AND/OR Does the Project potentially impact other areas where endangered species may be present through transboundary affects? >>	No The project does not have a negative impact on biodiversity and endangered species. No construction is foreseen, and no additional resource extraction will happen. No transboundary effects from the project can be expected since the focus is on the dissemination of the F3PA efficient cook stoves on household level and does not influence any resources which	N/A	N/A

	could have transboundary effects.		

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

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APPENDIX 3-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to Annex A of Principles and Requirements for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
1.1	7 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption